

On aspectual coercion in *before*-/*after*-clauses: Evidence from processing

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Abstract. Certain *before*-clauses and *after*-clauses are *prima facie* ambiguous between a strong reading and a weak reading, and two types of theories of these sentences have been proposed. Under-specification theories posit that the temporal clauses in question are not actually ambiguous: their only LF corresponds to the weak reading, with the strong reading falling out as a subcase. Rett’s (Sinn und Bedeutung 24, 201–218, 2020) ambiguity theory, by contrast, claims that the strong reading surfaces by default, with the weak reading only arising if we coerce the temporal clause. Under standard psycholinguistic assumptions, coercion incurs a processing cost, and hence, we argue, Rett’s (2020) theory makes processing predictions that under-specification theories do not. We test these predictions via four self-paced reading experiments and find moderate evidence that they are borne out.

1 Introduction

As originally observed by Anscombe (1964) and Heinämäki (1974), *before* and *after* exhibit two puzzling asymmetries in their truth-conditions. First, sentences with atelic predicates in *after*-clauses (1a) are *prima facie* ambiguous (i.e. they can be thought of as true in two types of scenarios), whereas analogous sentences with *before* (1b) are not (Anscombe 1964). To see the asymmetry, suppose that Hector was happy from 6pm–8pm and consider three scenarios for when Emma was happy (Scenario A, 5–5:30pm; Scenario B, 6:30–7pm; Scenario C, 8–8:30pm); see Figure 1 for a visual representation. (1a) is true if Emma was happy after Hector stopped being happy (Scenario C) or if she was happy after Hector started (Scenario B). By contrast, (1b) is only true if Emma was happy before Hector started being happy (Scenario A). (1b) lacks a “before-finish” reading: if Emma was happy before Hector stopped being happy but not before he started (Scenario B), (1b) is false.

- (1) a. Emma was happy after Hector was happy.
(✓after-start, ✓after-finish)
- b. Emma was happy before Hector was happy.
(✓before-start, ✗before-finish)

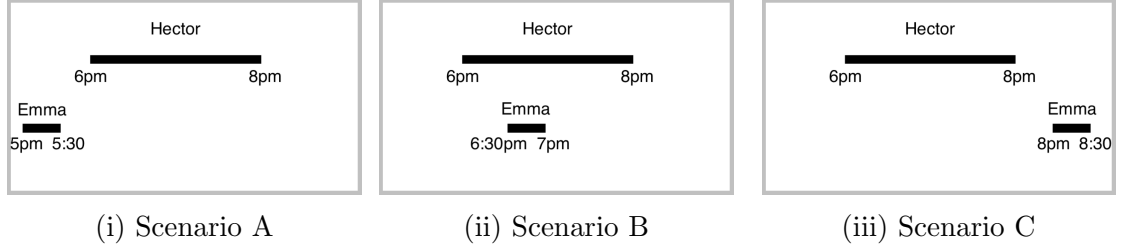


Figure 1: Supplemental timelines for (1).

Second, sentences with perfective accomplishments in *before*-clauses (2a) are *prima facie* ambiguous, whereas parallel sentences with *after* (2b) are not (Heinamäki 1974). To see this, suppose that Jackson climbed a mountain in twelve hours (9am-9pm) and consider three scenarios for when Emma was nervous (Scenario A, 5am-8pm; Scenario B, 12pm-3pm; Scenario C, 10pm-1am); see Figure 2. (2a) is true if Emma was nervous before Jackson started climbing the mountain (Scenario A) or if she was nervous before he finished (Scenario B). By contrast, (2b) is only true if Emma was nervous after Jackson finished (Scenario C). If Emma was nervous after he started but not after he finished (Scenario B), (2b) is false.

- (2) a. Emma was nervous before Jackson climbed the mountain.
 (✓before-start, ✓before-finish)
 b. Emma was nervous after Jackson climbed the mountain.
 (✗after-start, ✓after-finish)

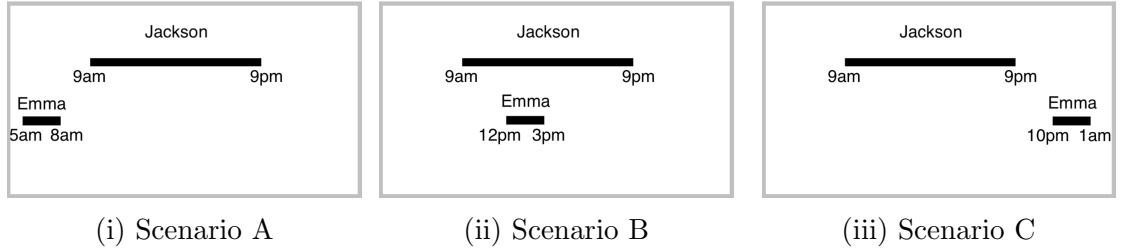


Figure 2: Supplemental timelines for (2).

There are two types of theories of these asymmetries between *before* and *after*, which we will call under-specification theories and ambiguity theories. Under-specification theories (Anscombe 1964; Heinamäki 1974; Beaver & Condoravdi 2003; Krifka 2010), observing that there is an entailment relation between the two “readings” of (1a) and between the two “readings” of (2a), posit that (1a) and (2a) are not actually ambiguous. These accounts use lexical entries for *before/after* that, when paired with independently motivated assumptions about telicity, derive the

correct truth-conditions for (1) and (2) without positing any ambiguities. On this sort of picture, (1a)’s only LF has weak truth-conditions (“Emma was happy sometime after Hector started being happy”), truth-conditions which are verified both by after-start contexts like Scenario B and by after-finish contexts like Scenario C. (2a) receives a similar treatment.

In ambiguity theories (Condoravdi 2010; Rett 2020), by contrast, the LF for one reading of (1a) and (2a) has a coercion operator that the other lacks. As an exemplar, consider Rett’s (2020) ambiguity theory. Rett (2020) proposes that in the absence of silent material, *p before q* has a before-start meaning (“*p* is true before the first time at which *q* is true”), and *p after q* has an after-finish meaning (“*p* is true after the last time at which *q* is true”). To account for (1a)’s after-start reading, Rett assumes that a silent **inchoative coercion** operator (Binnick 1991; Brennan & Pylkkänen 2010)—which we call INCHOAT—can attach to *Hector was happy* and coerce it into a change-of-state (*Hector started being happy*). In other words, *after Hector was happy* (for Rett) means “after Hector’s happiness ended,” while *after INCHOAT Hector was happy* means “after Hector’s happiness started.” To account for (2a)’s before-finish reading, Rett assumes that a silent **completive coercion** operator (which we call COMPLET) can attach to *Jackson climbed the mountain* and coerce it into an achievement (*Jackson finished climbing the mountain*). In other words, *before Jackson climbed the mountain* (for Rett) means “before Jackson started climbing the mountain,” whereas *before COMPLET Jackson climbed the mountain* means “before Jackson finished climbing the mountain.”

Rett (2020) claims that INCHOAT and COMPLET are not *ad hoc* but rather independently needed to deal with other data, such as (3). In (3a), an atelic predicate (*Hector was happy*) receives a change-of-state interpretation (*Hector started being happy*), which we can capture by assuming that the *within*-modifier forces INCHOAT to apply to *Hector was happy* (Binnick 1991; Brennan & Pylkkänen 2010). Regarding (3b), Rett (2020) notes that it has a reading paraphraseable as “Jackson finished climbing the mountain at 7pm sharp,” and she accounts for this by assuming that the *at*-adverbial forces COMPLET to apply to *Jackson climbed the mountain*.

- (3) a. Within an hour, Hector was happy.
- b. Jackson climbed the mountain at 7pm sharp.

The debate between under-specification and ambiguity theories remains unsettled. On the empirical side, much of the debate has centered around NPI-licensing facts, where both types of theories have advantages and disadvantages (see discussion in Rett 2020). On the conceptual side, Rett’s (2020) appeals to (3a-b) could be construed as evidence for her position if coercion-based theories of these cases are indeed correct, but the necessity of appealing to coercion in cases like (3a) has come into question (Krifka 2010). Additionally, while Krifka (2010) notes that a

coercion-less theory is the most parsimonious in the absence of strong evidence that sentences like (1a)/(2a) are genuinely ambiguous, Rett (2020) counters that these sentences are overtly ambiguous in some languages, with the two readings being distinguished via overt aspectual morphemes that resemble her covert operators.

The present study contributes to the debate between under-specification and ambiguity theories by testing several online processing predictions that Rett’s (2020) theory makes and under-specification theories do not. Rett’s (2020) theory makes distinctive online processing predictions because coercion has a distinctive psycholinguistic profile (Todorova et al. 2000; Husband et al. 2006; Brennan & Pykkänen 2008; Brennan & Pykkänen 2010; Lukassek et al. 2017). In particular, under standard assumptions, parsers do not insert coercion operators by default and thus show processing slowdowns in a coercion sentence once they know that coercion is required to resolve a mismatch with an adverbial or with contextual knowledge. Using this perspective on coercion, previous studies have tested coercion vs. non-coercion theories of various data, arguing for coercion if the claimed coercion cases have an online cost (Piñango et al. 1999, 2006; Todorova et al. 2000; McElree et al. 2001; Traxler et al. 2002, 2005; Brennan & Pykkänen 2008, 2010; Lukassek et al. 2017) and arguing against coercion otherwise (Pickering et al. 2006; Lukassek et al. 2017). Indeed, Brennan & Pykkänen (2010) have previously found a processing cost for (3a), which can be seen as providing indirect support for Rett’s proposal. At the same time, no prior experimental work has directly examined the prospect of coercion in *before-/after*-clauses or the notion of completive coercion in general.

Once we take past findings on the processing of coercion into account, we see that if Rett’s theory is correct, parsers who see (1a) in an after-start context or (2a) in a before-finish context—that is, parsers who see (1a)/(2a) in contexts where the only available reading is one that involves coercion—should show processing slowdowns upon realizing that coercion is necessary. Additionally, given Rett’s (2020) claim that (3a-b) also involve coercion, she predicts that they should exhibit coercion-related slowdowns too. Absent additional assumptions, under-specification theories do not predict that the processing of (1) and (2) should be affected by context (so long as the sentence is true) and are agnostic about (3a-b).

We test these predictions via four self-paced reading experiments. In Experiments 1a-b, we investigate whether sentences like (3a-b) have the processing profile predicted by coercion-based theories, finding that while (3b) is costly in same way as other coercion cases, (3a) does not incur a processing cost at all. The latter result constitutes a replication failure vis-à-vis Brennan & Pykkänen (2010).

After Experiment 1, we follow up on the completive and inchoative coercion results in turn. Starting with completive coercion: since Experiment 1b finds an online cost of COMPLET-insertion for (3b), Rett would expect that we should find a similar cost for (2a)’s before-finish reading. We test this prediction in Experiment 2

and find that the before-finish reading is indeed costly, though the online effect found in Experiment 2 is delayed by one word vis-à-vis the effect found in Experiment 1.

Turning to inchoative coercion, Experiment 3 addresses one possible reason why Experiment 1a did not replicate Brennan & Pylkkänen (2010), but we still fail to find a cost for (3a). Finally, Experiment 4 tests whether (1a)’s after-start reading has a coercion operator that the other reading lacks, finding that after-start readings of *after* have the same sort of delayed cost as before-finish readings of *before*.

Taken together, our results are compatible with Rett’s (2020) theory of (1), (2), and (3b), though the results of Experiment 1a and 3 suggest that using *within*-modifier sentences to motivate a coercion-based account of (1a) may be untenable. Additionally, the timing of the cost for after-start readings of *after* and before-start readings of *before* is *prima facie* unexpected under Rett’s theory, since past work on aspectual coercion has typically found slightly earlier slowdowns. In response to this worry, we argue that Rett’s theory can account for the timing of the cost in Experiments 2 and 4 by appealing to the fact that coercion in *before/after*-clauses is “pragmatic” (i.e. performed to resolve a mismatch between an adverbial and a context), whereas the cases of coercion subject to experimental scrutiny in the past have been “semantic” (i.e. performed to resolve a sentence-internal type mismatch). However, further research is needed to assess whether this explanation is tenable.

Although our results require Rett (2020) to make some additional assumptions, we argue that the results align better with Rett (2020) than with other existing theories of *before* and *after*, which do not (as it stands) predict any asymmetries of the sort observed in Experiments 2 and 4.

By providing moderate support for Rett’s idea that there are covert operators in *before/after*-clauses that do similar semantic work as overt morphemes in other languages, these results contribute to building a typology of covert aspectual operators. By extension, these results touch on the broader question of whether languages that lack a certain functional morpheme overtly actually have a covert operator that does similar semantic work; this question arises not just in the domain of aspect but also domains like definiteness and kind-reference (Chierchia 1998). In addition to these implications for formal semantic theory, our results have consequences for theories of processing as well: in particular, if our explanation of the delayed costs in Experiments 2 and 4 is on the right track, then that would support the idea of a lag between semantic and pragmatic processing (Huang & Snedeker 2009, a.o.)

The rest of this paper is structured as follows. In section 2, we provide further background on under-specification vs. ambiguity theories of *before/after* (section 2.1) and aspectual coercion (section 2.2). Section 3 lays out the predictions made by Rett’s (2020) theory in more detail. Sections 4-7 cover the methodology, results, and takeaways from Experiments 1-4. Section 8 consists of a general discussion of the results and their theoretical consequences.

2 Background

2.1 *Before and after*: Under-specification or ambiguity?

In this subsection, we expound a characteristic example of an under-specification theory of *before/after* (section 2.1.1) and contrast it with Rett 2020 (section 2.1.2). Although there are multiple under-specification theories, they do not differ in their processing predictions, so we do not discuss them separately here.¹

Throughout, we make four formal background assumptions. First, we assume that untensed sentences denote sets of time-intervals, type $\langle i, t \rangle$. Second, we make the simplifying assumption that there is one semantically interpreted tense operator in *before/after*-sentences. PAST, for example, takes a set of time-intervals p and asserts that p has a member that precedes the utterance time t_u (Prior 1958).

$$(4) \quad \llbracket \text{PAST} \rrbracket = \lambda p_{\langle i, t \rangle}. \exists t' < t_u: p(t') = 1$$

Third, we assume that an untensed stative like $\llbracket \text{Hector be happy} \rrbracket$ contains the maximal interval throughout which Hector was happy and its subintervals. For example, if Hector was happy 6-8pm, then $\llbracket \text{Hector be happy} \rrbracket$ contains $[6\text{pm}, 8\text{pm}]$ and its subintervals, all the way down to points. Finally, an accomplishment like $\llbracket \text{Jackson climb the mountain} \rrbracket$ will be a singleton containing the interval throughout which Jackson climbed the mountain. For example, if Jackson climbed the mountain in the span $[9\text{am}, 9\text{pm}]$, then $\llbracket \text{Jackson climb the mountain} \rrbracket = \{[9\text{am}, 9\text{pm}]\}$. These assumptions about lexical aspect are standard (Taylor 1977; Dowty 1979).²

2.1.1 An under-specification theory of *before/after* (Anscombe 1964)

As an exemplar of an under-specification approach, we present a version of Anscombe's (1964) classic theory, minimally modified so as to account for the data with accomplishments (which she does not consider). We illustrate the theory with (1) and (2), repeated below.

- (5) a. Emma was happy *before/after* Hector was happy.
 b. Emma was nervous *before/after* Jackson climbed the mountain.

(5a-b) receive the LFs in (6a-b), the derivations for which involve the modernized variants of Anscombe's (1964) entries in (7a-b). (7b) is equivalent to Anscombe's (1964) entry for *after*; the only difference between (7a) and Anscombe's (1964)

¹The only ambiguity theory other than Rett's is Condoravdi (2010), whose theory we return to in section 8.3. To preview, we will see that despite being an ambiguity theory, Condoravdi (2010) makes processing predictions that are closer to under-specification theories than to Rett (2020).

²Like all theories of *before/after* cited in this section, we ignore grammatical aspect, but we believe that one can incorporate it into the theory without issue.

original entry for *before* is that her entry invokes the relation $t'' > t'$ (“all subintervals of t'' follow t' ”) in the spot where (7a) invokes the relation $t'' >_{\exists} t$ (which we define to mean that some subintervals of t'' follow t).³ As detailed below, this weakening is necessary to make Anscombe’s (1964) theory work with accomplishments.

- (6) a. [PAST [[Emma be happy] [before/after [Hector be happy]]]]
- b. [PAST [[Emma be nervous] [before/after [Jackson climb the mountain]]]]
- (7) a. $\llbracket \text{before} \rrbracket = \lambda q_{\langle i, t \rangle} \cdot \lambda p_{\langle i, t \rangle} \cdot \lambda t'. p(t') = 1 \wedge \forall t'' [q(t'') = 1 \rightarrow t'' >_{\exists} t']$
- b. $\llbracket \text{after} \rrbracket = \lambda q_{\langle i, t \rangle} \cdot \lambda p_{\langle i, t \rangle} \cdot \lambda t'. p(t') = 1 \wedge \exists t'' [q(t'') = 1 \wedge t' > t'']$

In (6a-b), *before/after* combine with the embedded and matrix clauses in succession, and the result is fed to PAST. (8a) shows the truth-conditions for the version of (6a) with *before*, while (8b) shows the truth-conditions for the version with *after*.

- (8) a. $\llbracket [\text{PAST} [\llbracket \text{Emma be happy} \rrbracket [\text{before} [\llbracket \text{Hector be happy} \rrbracket]]]] \rrbracket = 1$ iff
 $\exists t' < t_u [t' \in \llbracket \text{Emma be happy} \rrbracket \wedge$
 $\forall t'' [t'' \in \llbracket \text{Hector be happy} \rrbracket \rightarrow t'' >_{\exists} t']]$
- b. $\llbracket [\text{PAST} [\llbracket \text{Emma be happy} \rrbracket [\text{after} [\llbracket \text{Hector be happy} \rrbracket]]]] \rrbracket = 1$ iff
 $\exists t' < t_u [t' \in \llbracket \text{Emma be happy} \rrbracket \wedge$
 $\exists t'' [t'' \in \llbracket \text{Hector be happy} \rrbracket \wedge t' > t'']]$

To see that these truth-conditions are right, consider the three scenarios in Figure 1 again. The theory correctly predicts that (8a) is only true in the before-start scenario (Scenario A), as that is the only scenario where we can find an Emma-be-happy time t' such that every Hector-be-happy time partially or fully follows t' . If one worries at this point about whether the use of $>_{\exists}$ (“partially or fully follows”) makes the truth-conditions in (8a) too weak, remember that since *Hector be happy* is stative, there are Hector-be-happy times that are points. Points do not have subparts, and hence, any given point either fully follows t' or does not follow it at all. So if every Hector-be-happy time partially or fully follows t' , this means that every point at which Hector is happy fully follows t' , corresponding to intuitions. Turning to (8b), the theory correctly predicts that it is true in Scenarios B and C: in both, there is an Emma-be-happy time t' such that there is some Hector-be-happy time before t' .

(5b) shows the predicted truth-conditions for our telic example.

- (9) a. $\llbracket [\text{PAST} [\llbracket \text{Emma be nervous} \rrbracket [\text{before} [\llbracket \text{Jackson climb the mountain} \rrbracket]]]] \rrbracket =$
 1 iff $\exists t' < t_u [t' \in \llbracket \text{Emma be nervous} \rrbracket \wedge$
 $\forall t'' [t'' \in \llbracket \text{Jackson climb the mountain} \rrbracket \rightarrow t'' >_{\exists} t']]$

³Formally, $t'' >_{\exists} t'$ iff $\exists t''' [t''' \sqsubseteq t'' \text{ and } t''' > t']$.

- b. $\llbracket \text{[PAST [Emma be nervous] [after [Jackson climb the mountain]]]} \rrbracket = 1$ iff $\exists t' < t_u$ $[t' \in \llbracket \text{Emma be nervous} \rrbracket] \wedge \exists t'' [t'' \in \llbracket \text{Jackson climb the mountain} \rrbracket \wedge t' > t'']$

To verify that these truth-conditions are correct, let us return to the three scenarios in Figure 2. Keeping in mind that $\llbracket \text{Jackson climb the mountain} \rrbracket$ denotes the singleton $\{[9\text{am}, 9\text{pm}]\}$ in all three scenarios, we see that (9a) comes out true in both Scenario A (the before-start scenario) and Scenario B (before-finish), as desired: in both, there is an Emma-be-nervous time t' such that some subintervals of $[9\text{am}, 9\text{pm}]$ follow t' . The theory also correctly predicts that (9b) is only true in the after-finish scenario (Scenario C), as that is the only scenario where Emma was nervous sometime after $[9\text{am}, 9\text{pm}]$.

Note that if we replaced $>_{\exists}$ in (8a) with $>$ (as in Anscombe's original theory), (8a) would come out true iff there is an Emma-be-nervous time that precedes $[9\text{am}, 9\text{pm}]$. These truth-conditions are too strong, as they are only verified by Scenario A.

2.1.2 Rett (2020)

Having presented an exemplar of an under-specification theory, we now contrast it with Rett 2020. Rett (2020)-style entries for *before/after* are given in (10a-b). These entries are slightly simplified relative to Rett's (2020) actual entries but are empirically equivalent for the data of interest.

- (10) a. $\llbracket \text{before} \rrbracket = \lambda q_{\langle i, t \rangle}. \lambda p_{\langle i, t \rangle}. \lambda t'. p(t') = 1 \wedge \forall t'' [q(t'') = 1 \rightarrow t' < t'']$
 b. $\llbracket \text{after} \rrbracket = \lambda q_{\langle i, t \rangle}. \lambda p_{\langle i, t \rangle}. \lambda t'. p(t') = 1 \wedge \forall t'' [q(t'') = 1 \rightarrow t' > t'']$

(11) shows the predicted truth-conditions for (5a) without silent material.

- (11) $\llbracket \text{[PAST [Emma be happy] [before/after [Hector be happy]]]} \rrbracket = 1$ iff $\exists t' < t_u$ $[t' \in \llbracket \text{Emma be happy} \rrbracket] \wedge \forall t'' [t'' \in \llbracket \text{Hector be happy} \rrbracket \rightarrow t' \llbracket < / > t'']$

While (11) gives the correct truth-conditions for *Emma was happy before Hector was happy*, the truth-conditions predicted for the analogous sentence with *after* are too strong. To see this, let us return to Figure 1 once again. (11) correctly predicts that *Emma was happy before Hector was happy* is only true in Scenario A (before-start), as that is the only scenario where there is an Emma-be-happy time that precedes every time at which Hector was happy. The analogous prediction for *Emma was happy after Hector was happy* is that it should only be true in Scenario C (the scenario where there is an Emma-be-happy time that follows every Hector-be-happy time), but it is intuitively true in Scenario B as well.

To capture the full range of readings, Rett (2020) posits that whenever we judge *Emma was happy after Hector was happy* true in an after-start scenario like Scenario

B, we are coercing *Hector was happy* into having a change-of-state interpretation (*Hector started being happy*). Following much literature on aspectual coercion (see section 2.2), Rett assumes that the process of coercion involves inserting a silent operator into the structure. For the case at hand, Rett (2020) proposes that the relevant operator is an **inchoative coercion** operator (Binnick 1991; Dölling 2014, a.o.), defined below. Note that $t \sqsubseteq t'$ means “ t is a subinterval of t' .”

$$(12) \quad \text{INCHOAT}(p) = \{t. p(t) = 1 \wedge \forall t' [p(t') = 1 \rightarrow [t < t' \vee t \sqsubseteq t']]\}$$

In words, $\text{INCHOAT}(p)$ —which Rett (2020) assumes is defined only if p is atelic—returns the singleton set containing the first point at which p holds. For example, if Hector was happy from 6-8pm, then $\text{INCHOAT}(\llbracket \text{Hector be happy} \rrbracket) = \{6\text{pm}\}$.

(13a) shows the truth-conditions for *Emma was happy after INCHOAT Hector was happy* in our example where Hector was happy 6pm-8pm. While the version of the sentence without INCHOAT is only true in an after-finish scenario like Scenario C, (13a) is also true in Scenario B. INCHOAT can in principle attach within an atelic *before*-clause as well (13b), but it would be vacuous: (13b), just like the version of the sentence without INCHOAT , is only true in Scenario A (the before-start scenario).

$$(13) \quad \begin{aligned} \text{a. } & \llbracket [\text{PAST } \llbracket \text{Emma be happy} \rrbracket \text{ [after } [\text{INCHOAT } \llbracket \text{Hector be happy} \rrbracket]]] \rrbracket = \\ & 1 \text{ iff } \exists t' < t_u [t' \in \llbracket \text{Emma be happy} \rrbracket \wedge \forall t'' [t'' \in \{6\text{pm}\} \rightarrow t' > t'']] \\ \text{b. } & \llbracket [\text{PAST } \llbracket \text{Emma be happy} \rrbracket \text{ [before } [\text{INCHOAT } \llbracket \text{Hector be happy} \rrbracket]]] \rrbracket = \\ & 1 \text{ iff } \exists t' < t_u [t' \in \llbracket \text{Emma be happy} \rrbracket \wedge \forall t'' [t'' \in \{6\text{pm}\} \rightarrow t' < t'']] \end{aligned}$$

Turning to our telic example, (14) shows the predicted truth-conditions without silent material.

$$(14) \quad \llbracket [\text{PAST } \llbracket \text{Emma be nervous} \rrbracket \llbracket \boxed{\text{before}} \text{ / after } \llbracket \text{Jackson climb the mountain} \rrbracket]] \rrbracket = 1 \text{ iff } \exists t' < t_u [t' \in \llbracket \text{Emma be nervous} \rrbracket \wedge \forall t'' [t'' \in \llbracket \text{Jackson climb the mountain} \rrbracket \rightarrow t' \boxed{</>} t'']]$$

While (14) gives the right meaning for *Emma was nervous after Jackson climbed the mountain*, the meaning predicted for the analogous *before*-sentence is too strong. To see this, consider Figure 2 one more time. (14) correctly predicts that *Emma was nervous after Jackson climbed the mountain* is only true in the after-finish Scenario C, as that is the only scenario where Emma was nervous sometime after [9am, 9pm], the runtime of Jackson climbing the mountain. The parallel prediction for *Emma was happy before Jackson climbed the mountain* is that it should only be true in the before-start Scenario A, but it is in fact true in Scenario B too.

To capture the full range of readings, Rett (2020) posits that whenever we judge *Emma was nervous before Jackson climbed the mountain* true in a before-finish scenario like Scenario B, we are coercing *Jackson climbed the mountain* into

an achievement (*Jackson finished climbing the mountain*). Rett (2020) formalizes this idea with the help of a **completive coercion** operator, defined below.⁴

$$(15) \quad \text{COMPLET}(p) = \{\iota t. \exists t' [p(t') = 1 \wedge t \sqsubseteq t' \wedge \forall t'' [t'' \sqsubseteq t' \rightarrow [t > t'' \vee t \sqsubseteq t'']]\}$$

In words, $\text{COMPLET}(p)$ —which Rett (2020) assumes is defined only if p is telic—finds the latest point contained within p ’s unique member and returns the singleton set containing that point.⁵ For example, if Jackson climbed the mountain from 9am–9pm, $\text{COMPLET}(\llbracket \text{Jackson climb the mountain} \rrbracket) = \{9\text{pm}\}$.

(16a) shows the predicted truth-conditions for *Emma was nervous before COMPLET Jackson climbed the mountain* in our example where Jackson climbed the mountain from 9am–9pm. While the version of the sentence without COMPLET was only true in Scenario A, (16a) is also true in Scenario B. COMPLET can in principle attach within a telic *after*-clause as well (16b), but it would be vacuous: (16b), just like the version of the sentence without COMPLET , is only true in Scenario C.

- (16) a. $\llbracket [\text{PAST} [\llbracket \text{Emma be nervous} \rrbracket [\text{before} [\text{COMPLET} [\text{J. climb the mountain}]]]] \rrbracket$
 $= 1 \text{ iff } \exists t' < t_u [t' \in \llbracket \text{Emma be nervous} \rrbracket \wedge \forall t'' [t'' \in \{9\text{pm}\} \rightarrow t' < t'']]$
 b. $\llbracket [\text{PAST} [\llbracket \text{Emma be nervous} \rrbracket [\text{after} [\text{COMPLET} [\text{J. climb the mountain}]]]] \rrbracket$
 $= 1 \text{ iff } \exists t' < t_u [t' \in \llbracket \text{Emma be nervous} \rrbracket \wedge \forall t'' [t'' \in \{9\text{pm}\} \rightarrow t' > t'']]$

Rett (2020) provides several motivations for her approach. First, she observes (based on a survey of 17 languages) that all languages seem to allow before-start readings of *before* and after-finish readings of *after*, but not all languages allow before-finish readings of *before* or after-start readings of *after*. This fact arguably speaks in favor of a theory where before-start readings of *before* and after-finish readings of *after* are “basic” and other readings are derived via an additional language-specific mechanism. Second, Rett (2020) notes that languages like Tagalog distinguish before-start and before-finish readings using overt aspectual morphology akin to COMPLET . In Tagalog, before-start readings use the “neutral” perfective (17a), whereas before-finish readings use the “ability and involuntary action” perfective (Dell 1983), which is often translated as “manage to” (17b).⁶ Rett (2020) also notes that many languages use an inchoative marker in atelic *after*-clauses, but she does not find a language where the inchoative marker changes the interpretation.

⁴Rett (2020) cites Dölling (2014) for the notion of completive coercion, but as far as we can see, Rett’s (2020) notion of completive coercion is different from Dölling’s.

⁵In (15), p is required to be a singleton because otherwise, there will not be a unique t meeting the description. Given this, we would need to tweak (15) to account for cases with non-singleton accomplishments (*before Jackson climbed a mountain*), but for simplicity, we stick with (15).

⁶The examples in (17a-b) come from Rett’s (2020) own fieldwork, glosses slightly simplified. PFV.NEUT = neutral perfective, AIA = ability and involuntary action.

- (17) a. Um-alis siya bago niya w<in>alisan angсахиг
 PFV.NEUT-leave 3SG before 3SG <PFV.NEUT>-sweep floor
 ‘She left before he swept the floor.’ (before-start reading only)
- b. Um-alis siya bago niya **na**-walisan angсахиг
 PFV.NEUT-leave 3SG before 3SG AIA-sweep floor
 ‘She left before he swept the floor.’ (before-finish)

Third, Rett (2020) emphasizes that INCHOAT and COMPLET may be required on independent grounds. INCHOAT has been invoked for a variety of other data (Binnick 1991; Brennan & Pylkkänen 2010; Dölling 2014), including sentences where an atelic predicate occurs alongside a *within*-modifier (18a). These sentences have a change-of-state interpretation (*within an hour Hector became happy*), which is expected if we assume that INCHOAT obligatorily applies to *Hector be happy* in (18a). To motivate COMPLET, Rett (2020) notes that sentences like (18b) can have a “completive” reading (*Jackson finished climbing the mountain at 7pm*) and accounts for this by assuming that COMPLET obligatorily applies to *Jackson climbed the mountain* in (18b).

- (18) a. Within an hour Hector was happy.
 b. Jackson climbed the mountain at 7pm sharp.

Taken together, these arguments provide reasons for thinking that an ambiguity theory may have conceptual advantages (though see Krifka 2010 for the conceptual advantages of under-specification theories). However, there is a dearth of English-internal empirical arguments for the claim that *before/after* clauses, on certain readings, have covert morphemes analogous to aspectual markers in other languages; hence, the debate between under-specification and ambiguity theories is not fully settled. This paper contributes to the debate by identifying and testing some distinctive processing predictions made by Rett (2020), predictions that become clear upon taking a closer look at the literature on coercion phenomena.

2.2 Past literature on coercion

There are a variety of data for which coercion-based theories have been proposed. In addition to the comparatively little-studied cases discussed so far (to which we return in section 3), two particularly well-studied cases involve sentences like (19a) and (19b). In (19a), an entity-denoting expression (*the book*) takes on an eventive meaning (‘reading the book’) in the presence of a verb like *finish*; in (19b), a punctual predicate (*jump*) takes on a repetitive meaning (*jump repeatedly*) in the presence of a *for*-adverbial. See Bott (2010) and Dölling (2014) for even more examples of potential coercion phenomena.

- (19) a. I finished the book that you told me to read.
 b. For an hour, the zebra jumped in its enclosure.

Proponents of coercion-based theories generally assume that sentences with coercion contain a covert operator that is inserted to repair an otherwise ill-formed sentence (following Moens & Steedman 1988, a.o.). In some cases, the ill-formedness in question is a compositional mismatch. For example, proponents of a coercion theory of (19a) assume that there is a type mismatch between certain verbs (*begin*, *finish*, etc.) and entity-denoting expressions like *the book*; to resolve the mismatch, we insert a **complement coercion** operator, which coerces *the book* into having an eventive meaning (Jackendoff 1997, McElree et al. 2001, Traxler et al. 2002, Traxler et al. 2005, and much other work, *pace* Piñango & Deo 2016). In other cases, the proposed ill-formedness that motivates coercion can either be thought of as a compositional mismatch or as a pragmatic/world knowledge violation. For example, some proponents of a coercion theory of (19b) posit that there is a compositional mismatch between the durative adverbial *for an hour* and the punctual predicate *jump* (de Swart 1998). Others suggest that the semantic derivation for (19b) is mismatch-free, but the resulting meaning (“the zebra jumped once, and it lasted for an hour”) is ill-formed because it conflicts with world knowledge (Dölling 2014). Both variants of the coercion theory propose that (19b) is repaired by insertion of an **iterative coercion** operator, i.e. a silent “repeatedly.”

There are yet other claimed coercion cases whose underlying ill-formedness can only be thought of as pragmatic in nature. Consider (20), from Dölling (2014).

- (20) Mike swam for twenty years.

Dölling (2014) observes that in many contexts, the atelic predicate *swam* in (20) will receive the habitual interpretation “was a swimmer” rather than a literal one. It would be implausible to posit a compositional mismatch between *swim* and *for*-adverbials (which are prototypical modifiers of atelic predicates). Rather, Dölling (2014) suggests, the habitual reading arises because the literal meaning of (20), “Mike swam non-stop for twenty years straight,” conflicts with world knowledge, necessitating insertion of a habitual operator.

Coercion theories are often contrasted with under-specification theories (Brennan & Pylkkänen 2008; Lukassek et al. 2017). To illustrate the difference, let us consider (19b) again. While coercion theories of (19b) posit that *jump* is specified for a punctual meaning (“jump once”) and thus conflicts with the durative adverbial *for an hour*, an under-specification theory would say that *jump* is under-specified in the sense that it is equally compatible with the punctual meaning “jump once” and the durative meaning “jump repeatedly” (Brennan & Pylkkänen 2008). On the under-specification view, (19b) is not ill-formed in any way; rather, the

for-adverbial simply resolves the under-specification in favor of the “jump repeatedly” meaning. Other adverbials, such as *at*-adverbials (*The zebra jumped at 9pm*), resolve the under-specification in the other direction (“jump once”).

Coercion theories make processing predictions that under-specification theories do not (Todorova et al. 2000; Brennan & Pylkkänen 2008; Lukassek et al. 2017). If a sentence is underlyingly ill-formed and requires emergency insertion of a covert operator, we expect that parsers encountering such sentences will show processing slowdowns upon realizing that insertion of the operator is necessary; these slowdowns reflect the cognitive operations involved in inserting the operator into the structure. This prediction of coercion accounts holds regardless of whether the ill-formedness that necessitates operator insertion is semantic or pragmatic (Brennan & Pylkkänen 2008). By contrast, there is no reason to expect that resolving under-specification in one direction would be more costly than resolving it in another.

Because coercion and under-specification theories make distinct processing predictions, real-time processing studies have been an important testbed for coercion vs. under-specification theories of various phenomena. The majority of such studies look at claimed cases of complement coercion or iterative coercion. The theory of complement coercion predicts that sentences like (21a) should be hard to process at or shortly after *book*, as parsers have seen *begin* + an entity-denoting complement at that point and thus know that complement coercion is required. All studies on (21a) known to us (starting with McElree et al. 2001) bear out this prediction of the complement coercion account, finding longer processing times at or one word after *book* in (21a) compared to the non-coercion sentence (21b).

- (21) a. The children began the book in the afternoon.
- b. The children read the book in the afternoon.

Similarly, the theory of iterative coercion predicts that sentences like (19b) should be harder to process than controls at or one word after *jumped*, as parsers have seen *for an hour* + a punctual predicate and thus know that coercion is required. Most studies bear out this prediction and thus conclude in favor of the coercion approach (Piñango et al. 1999, 2006; Todorova et al. 2000; Husband et al. 2006; Brennan & Pylkkänen 2008). Conversely, Pickering et al. (2006) do not find a processing cost and accordingly argue for under-specification.

3 The processing predictions of Rett’s (2020) theory

In this section, we put Rett’s (2020) theory in conversation with the broader coercion literature by (I) reframing Rett’s (2020) invocations of coercion using the language of “ill-formedness and repair” that permeates the coercion literature; (II) contrasting the processing predictions of Rett’s theory with those of under-specification

theories; and (III) discussing the one previous study relevant to testing these predictions: Brennan & Pylkkänen 2010.

Following past literature (e.g. Binnick 1991, Brennan & Pylkkänen 2010), we can frame the coercion account of (22) in terms of a compositional mismatch. In particular, the claim is that *within*-modifiers select for telic predicates and thus clash compositionally with atelic predicates. Insertion of INCHOAT repairs the mismatch because INCHOAT(⟦Jessica was mad in office hours⟧), which has the change-of-state meaning “Jessica became mad in office hours,” is telic.

(22) Within an hour, Jessica was mad in office hours.

This account predicts that (22) should be hard to process at or one word after *was*, as parsers have seen a *within*-modifier + the beginning of a stative and thus know that INCHOAT is required.⁷ **Experiment 1a** tests this prediction, and **Experiment 3** tests a parallel prediction for subject-experiencer verbs (*within an hour, Ben loved Sue*), the types of predicates used by Brennan & Pylkkänen (2010).

One can reframe Rett’s (2020) completive coercion account of sentences like (23) either in terms of a compositional mismatch or in terms of a pragmatic violation. If one opts for the semantic route, one would say that there is a compositional mismatch between the durative VP *climbed the mountain* and the punctual adverbial *at 7pm sharp*. If one opts for the pragmatic route, one would say that the semantic derivation for (23) is mismatch-free but that the resulting meaning (“at 7pm sharp, Jackson both started and finished his climb”) conflicts with world knowledge. In either case, insertion of COMPLET resolves the mismatch because COMPLET(⟦Jackson climbed the mountain⟧), which has the meaning “Jackson finished climbing the mountain,” is punctual rather than durative.

(23) At 7pm sharp, Jackson climbed the mountain.

Both the semantic and pragmatic variants of the theory predict that in a context where Jackson climbed one mountain and the climb ended at 7pm, (23) should be hard to process at or one word after *climbed*; at that point, the parser has seen *at 7pm*, has seen the start of an accomplishment, and knows the climb ended at 7pm, so they know that COMPLET is needed.⁸ **Experiment 1b** tests this prediction.

We cannot reframe Rett’s (2020) theory of (24) in terms of a compositional mismatch, as the derivation for (24) does not crash without completive coercion,

⁷To elaborate: parsers who have read up to *was* in (22) will either predict a progressive, a predicative AP/PP, or a passive. We do not think that parsers in Experiment 1a below would predict a passive, and all other possible continuations, including progressive constructions, receive an inchoative interpretation when next to a *within*-modifier (*within an hour, he was helping her*).

⁸Since *climbed* can have atelic continuations (*climbed mountains*) one might think that parsers will not know the VP is telic in (23) until they read *the*. However, Jackson only climbed one mountain in the context, so the atelic continuation is implausible.

according to her: the coercion-less derivation merely expresses a different reading than the derivation with completive coercion. However, it is possible to recast her theory of (24) in pragmatic terms if we make the assumption that insertion of COMPLET is only licensed when the before-start reading would conflict with contextual information. In other words, when COMPLET is inserted in (24), it is inserted to repair a mismatch between the “default,” before-start truth-conditions of (24) and the context. This pragmatic assumption is in the spirit of Rett’s (2020) discussion: at one point, she suggests that insertion of COMPLET and INCHOAT might be conditioned by contextual factors, though she does not elaborate.

(24) Emma was nervous before Jackson climbed the mountain.

Under this theory, we would expect that in a context that is only compatible with the before-finish reading (and in which Jackson climbed one mountain), (24) should be hard to process at or one word after *climbed*. At that point, parsers have seen the start of an accomplishment in a *before*-clause but know that only a before-finish reading is compatible with the context; they thus spot the mismatch between (24)’s default truth-conditions and the context and know to insert COMPLET. Crucially, reading times at or one word after *climbed* in (24) should be longer in a before-finish context than in a before-start context, since coercion in (24) is not necessary in a before-start context. **Experiment 2** tests this prediction.

We can reframe Rett’s (2020) theory of atelic *after*-sentences like (25) in a similar fashion. In particular, we can assume that the after-finish reading is the only reading available in a neutral context, and insertion of INCHOAT is only licensed when the after-finish reading would conflict with contextual information.

(25) Dave was regretful after Lara feared the large dog.

Under this theory, we would expect that in a context that is only compatible with the after-start reading, (24) should be hard to process at or one word after *feared*; at that point, parsers will project an stative predicate but know that only an after-start reading is compatible with the context, so they know to coerce. **Experiment 4** tests this prediction.

Under-specification theories do not share Rett’s (2020) local processing predictions about (24) and (25). Absent additional assumptions, such theories do not predict that the processing of (24) and (25) should be affected by context (so long as the sentence is true); they are agnostic about (22) and (23).

We conclude this discussion of Rett’s (2020) predictions with a caveat. Past research has found online costs for coercions that repair a semantic violation (e.g. complement coercion) and for coercions that repair a violation that can either be thought of as semantic or pragmatic (e.g. iterative coercion). No psycholinguistic research known to us looks at potential cases of coercion whose proposed ill-formedness must be thought of as pragmatic (e.g. (24)-(25) or the habitual reading

of *Mike swam for twenty years*, see above), though Brennan & Pylkkänen (2008) provide neural imaging evidence suggesting that the underlying ill-formedness of iterative coercion sentences is pragmatic. As such, it remains unknown whether semantic and pragmatic coercion have the same psycholinguistic profile. However, past literature has generally assumed that they should incur a similar cost in behavioral paradigms like self-paced reading (Brennan & Pylkkänen 2008), so the *prima facie* predictions of Rett’s (2020) theory are best stated as above.

3.1 Brennan & Pylkkänen (2010)

While Rett’s (2020) predictions about (23), (24), and (25) have not been addressed by previous psycholinguistic research, there is one previous paper on the processing of cases like (22): Brennan & Pylkkänen (2010). Brennan & Pylkkänen (2010) are interested in the processing of multiple phenomena, but what is relevant for our purposes is that among other things, they compare the processing of claimed inchoative coercion cases like (26a) to non-coercive controls like (26b).

- (26) a. Within 1 week the dragon detested the town’s people.
- b. Obviously the dragon detested the town’s people.

All of Brennan & Pylkkänen’s (2010) critical pairs used subject-experiencer verbs. Their inchoative coercion sentences all differed from controls in terms of the sentence-initial adverbial, with the controls using a variety of different adverbials.

For the reasons detailed above, the theory of inchoative coercion would expect longer processing times at or just after the main verb in (26a) vis-à-vis (26b). In Brennan & Pylkkänen’s (2010) self-paced reading study (Experiment 1), this prediction was borne out. Their Experiment 1 also had an acceptability rating component, in which they found that sentences like (26a), like some other cases of coercion (Todorova et al. 2000; Lukasek et al. 2017), receive lower acceptability ratings than controls. In their Experiment 2, Brennan & Pylkkänen (2010) found that sentences like (26a) have the same magnetoencephalography (MEG) signature as other coercions, such as iterative coercion (Brennan & Pylkkänen 2008).

4 Experiment 1

We started testing the processing predictions of Rett’s (2020) theory by trying to replicate Brennan & Pylkkänen’s (2010) Experiment 1 and, in parallel, examining whether sentences with *at*-modifiers + accomplishments have the processing profile predicted by the completive coercion account. Collectively, these experimental manipulations test the validity of Rett’s claim that INCHOAT and COMPLET are independently needed outside of *before/after*-clauses.

Note that all of our experiments were pre-registered (Experiment 1: [click here](#), Experiment 2: [click here](#), Experiment 3: [click here](#), Experiment 4: [click here](#)). Data, materials, and analysis scripts can be found [here](#).

4.1 Methodology

Experiment 1 had two between-subjects sub-experiments, Experiments 1a and 1b, which focused on inchoative and completive coercion, respectively.

4.1.1 Participants

We recruited 220 self-reported native English speakers from Prolific (110 per sub-experiment), intending to use the subset of participants who met our inclusion criteria (see sec. 4.2) as our analyzed sample. All participants received £2.10.

For visualizations and analyses on the self-paced reading results, we excluded data from participants who gave a rating of 3 out of 5 or higher on at least three out of our six ungrammatical/false fillers; see below for details on these trial types. 27 participants in Experiment 1a and 31 participants in Experiment 1b were excluded under this criterion.⁹ In each sub-experiment, we also excluded reading times (RTs) that were more than three standard deviations from the mean for a given word type (following Brennan & Pylkkänen 2010); see Supplementary Materials for details.¹⁰

4.1.2 Materials

Experiment 1a. Experiment 1a was a conceptual replication of Brennan & Pylkkänen’s (2010) Experiment 1. Participants saw 26 trials, each of which proceeded in three phases. First, participants read a short context sentence-by-sentence. Then, they read a target sentence word-by-word using a self-paced reading interface. Finally, participants rated the naturalness of the target sentence given the context on a 5-point Likert scale.¹¹ Following Brennan & Pylkkänen (2010), naturalness

⁹This exclusion rate is quite high, especially since our exclusion criterion does not seem very strict. Despite the high exclusion rate, we will not arbitrarily tweak our exclusion criterion *post hoc*. One possible explanation for the high exclusion rate is that Likert ratings are just not an ideal way to gauge attention, as even attentive participants might not all use a Likert scale in the same way. In fact, Brennan & Pylkkänen (2010) (who had a much lower exclusion rate) used binary acceptability judgments rather than Likert ratings. Given this, we tried switching to binary judgments for Experiment 2’s attention checks (see section 5.2).

¹⁰Note that we pre-registered a slightly different criterion for excluding individual RTs: namely, excluding all RTs that are <150ms or >2500ms. However, we ultimately decided that it would be less arbitrary to use this Brennan & Pylkkänen (2010)-style criterion.

¹¹Each trial unfolded next to a thematically related picture (e.g. a stylized picture of an office with chairs for trials in Table 1).

ratings were included because they allowed us to construct attention checks (see sec. 4.2), and because they can provide further insight into participants’ behavior.

Experiment 1a had four critical trial types (see **Table 1** for a sample item set). On inchoative-coercion trials (e.g. (a) in Table 1), participants read a context like (27a), followed by a target sentence with an atelic predicate + *within*-adverbial. On inchoative-control trials (e.g. (b) in Table 1), participants read a context like (27b), followed by a target sentence with an atelic predicate + *for*-adverbial. Trials like (b) are suitable non-coercive controls for (a) because *for*-adverbials are prototypical modifiers of atelic predicates, and as such no theory predicts that (b) should be difficult. Note that inchoative-coercion and inchoative-control trials used different contexts because we felt that, e.g., trying to construct a single context that verifies the target sentences in both (a) and (b) would result in something too convoluted.

In addition to trials like (a) and (b), we also included trials with complement-coercion target sentences (e.g. (c) in Table 1) and matched controls with non-coercive verbs (d). These two trial types were included as a sanity check: replicating past findings that (c)’s target sentence is costly (see section 2.2) helps show that our paradigm is sensitive enough to capture coercion effects. If we replicate past findings on complement coercion but find no comparable effect for another coercion type in a given experiment, the null result on the second coercion type cannot be chalked up to an overall inability to capture coercion effects within our paradigm.

- (27) a. **Context A:** One semester, Jessica took a class taught by a professor, Alex, whose lectures were extremely difficult to understand. Jessica had heard that Alex was more coherent in office hours, so she scheduled a 30 minute appointment with him. Alex did nothing but read one of his own articles to Jessica in full; by the fifteen-minute mark, Jessica voiced her annoyance about the professor’s behavior, but he paid her no mind and kept reading.
- b. **Context B:** Same first two sentences as Context A, then: Alex did nothing but read one of his own articles to Jessica in full during the first fifteen minutes, which annoyed her the whole time; she tried to get Alex to stop, but he paid her no mind and kept reading until the fifteen-minute mark.

Trial type		Context	Target sentence
Inchoative	(a) Coercion	A	Within fifteen minutes Jessica was mad in office hours.
	(b) Control	B	For fifteen minutes Jessica was mad in office hours.
Complement	(c) Coercion	A/B	Alex continued his article despite Jessica’s protest.
	(d) Control	A/B	Alex read his article despite Jessica’s protest.

Table 1: Sample stimuli for Experiment 1a.

Three differences between Experiment 1a’s design and Brennan & Pykkänen (2010) bear mention. First, we departed from Brennan & Pykkänen (2010) in our use of contexts. This was because most of our experiments needed contextual support to single out the relevant reading, and we wanted to minimize differences across experiments. Second, Brennan & Pykkänen (2010) used a variety of adverbials in their inchoative-control trials (see section 3.1), whereas all of our inchoative-control trials used *for*-adverbials. We made this choice because the contrast between *within*-adverbials and *for*-adverbials is as minimal as possible and because Brennan & Pykkänen (2010) report some noise in their data that may have resulted from the different adverbials. Second, Brennan & Pykkänen’s (2010) inchoative coercion sentences exclusively used subject experiencer constructions, while Experiment 1a used *be*-statives. The theory of inchoative coercion does not expect a difference between these two constructions in terms of the online cost of applying INCHOAT to them; Brennan & Pykkänen (2010) only used subject experiencer verbs because they had an additional subject- vs. object-experiencer manipulation. we opted to switch to *be*-statives because it was slightly easier to create contexts for these predicates.

In addition to the four critical trial types, Experiment 1a had two types of fillers: ungrammatical fillers, which used target sentences with violations of selectional restrictions (e.g. *Karen meowed the phone close to midnight*), and grammatical fillers, some of which used target sentences that were true given the context and some of which used target sentences that were false.

Experiment 1b. The paradigm and structure of Experiment 1b were the same as Experiment 1a, but the target sentences and contexts differed. Experiment 1b had four critical trial types (see **Table 2**) and two slightly different types of contexts: those ending in an *at*-adverbial (e.g. ending A in (28)) and those ending in a phrase like *X minutes after...* (ending B). On completive-coercion trials (e.g. (a) in Table 2), participants read a context that ended in an *at*-adverbial, followed by a target sentence with an accomplishment + *at*-adverbial. On completive-control trials (e.g. (b) in Table 2), participants read a context that ended in *X minutes after...*, followed by a target sentence with an accomplishment + *in*-adverbial. Trials like (b) are suitable non-coercive controls because *in*-adverbials are prototypical modifiers of accomplishments. In addition to trials like (a) and (b), we also included complement-coercion trials (c) and matched controls (d) as a sanity check. The contexts for these trials ended with a combination of ending A and ending B, e.g. *...at 9pm, five minutes after Hector made the bet*.

- (28) Context: Hector is a lifelong hiker and his daughter Emma wants to learn more about the outdoors, so they went on a hiking trip. On the first night, Hector bragged about how quickly he could set up a tent, and Emma bet

him that he couldn’t do it in ten minutes. Hector started building the tent at 8:55pm, and the tent was fully set up (**A**) at 9pm. (**B**) five minutes after Emma made the bet.

Trial type		Context	Target sentence
Completive	(a) Coercion	A	At 9pm sharp Hector built the humble tent.
	(b) Control	B	In five minutes Hector built the humble tent.
Complement	(c) Coercion	A + B	Hector finished the tent in front of Emma.
	(d) Control	A + B	Hector built the tent in front of Emma.

Table 2: Sample stimuli for Experiment 1b.

The contexts here ensured that participants interpreted (a) with completive coercion (*at 9pm, Hector finished*) rather than in some other way, e.g. with a world knowledge violation (*at 9pm, Hector both started and finished*).¹²

We implemented the task in PCIBex (Schwarz & Zehr 2021). Sample trials from Experiments 1a and 1b are available [here](#) and [here](#), respectively.

4.1.3 Procedure

After providing consent, participants saw an instruction text and a practice trial (accessible [here](#)), which was the same regardless of whether the participant had been recruited for Experiment 1a or 1b.

Upon completing the practice trial, participants in Experiment 1a were sorted into one of eight lists, each of which contained 16 critical trials and 10 fillers. The 10 fillers were the same across lists and consisted of 4 ungrammatical fillers, 4 grammatical fillers that were true, and 2 grammatical fillers that were false. For the critical trials, we created 16 item sets like the item set in Table 1 (each using a different *be*-stative) and used a Latin square design where each participant saw one trial from each item set. For example, participants in lists A and B saw the inchoative-coercion trial from Table 1, participants in lists C-D saw the inchoative-control trial from this item set (b), participants in list E-F saw the complement-coercion trial (c), and participants in lists G-H saw the complement-control trial (d). As a result of this Latin square design, each participant saw 4 examples of each of the 4 critical trial types.

¹²While we could have used, e.g., a combination of endings A and B (*at 9pm, five minutes after Emma made the bet*) on all trial types, this might have made the first few words of (b)’s target sentence easier to process than (a)’s (because *five minutes* was read more recently than *at 9pm*), and we wanted to minimize the likelihood of pre-verbal differences between (a) and (b). The decision to use two kinds of contexts in Experiment 1b was made shortly prior to data collection; hence, we forgot to update the pre-registration to reflect this aspect of the final design.

The lists in Experiment 1a also differed as to whether complement-coercion trials (and controls) used *within*-contexts like (27a) or *for*-contexts like (27b). Complement-coercion trials (and controls) in lists B, D, F, and H used *within*-contexts, while lists A, C, E, and G used *for*-contexts on these trials.

Participants in Experiment 1b were sorted into one of four lists, each of which had the same trial breakdown as Experiment 1a. As in Experiment 1a, the 10 fillers were the same across lists; for the critical trials, we created 16 items sets like the one in Table 2 (each centered around a different accomplishment) and used a Latin square design. Note that Experiment 1b contains fewer lists overall because it lacked a context manipulation on complement-coercion trials and controls. In all of our experiments, trials were presented in fully random order.

4.2 Results

4.2.1 Experiment 1a

Figure 3 summarizes RT results for the analyzed subset of Experiment 1a’s data; note that we opted to use log-transformed RTs rather than raw RTs as the dependent variable in all of our experiments, as the former were more normally distributed. Figure 3’s left panel displays mean logRTs for complement coercion trials like (c) in Table 1 and matched controls like (d) in and around the region of interest for complement coercion. Recall that following past literature, we expect longer RTs for complement coercion trials relative to matched controls on either the direct object noun (the critical word) or noun+1. Figure 1’s right panel displays mean logRTs for inchoative coercion trials (e.g. (a) in Table 1) and matched controls like (b) around the region of interest for inchoative coercion. Recall that the INCHOAT-based theory expects longer RTs for inchoative coercion trials relative to controls at verb or verb+1.

We tested our RT hypotheses using a pair of mixed-effects linear regression models with a structure similar to the models used by Brennan & Pykkänen (2010) in their Experiment 1. We fit our models using the *lmerTest* package (Kuznetsova et al. 2017) in R (R Core Team 2025), an extension of the *lme4* package (Bates et al. 2015). Both models used data from all four critical trial types, with the first model being fit on critical word logRTs (verb in trials like (a-b) in Table 1; direct object noun in trials like (c-d)), and the other model being fit on critical+1 RTs (verb+1 in (a-b); noun+1 in (c-d)). The models predicted logRT from trial type and three covariates: word length (in characters), log lexical frequency according to the English Lexicon Project (Balota et al. 2007), and logRT on the previous word. These covariates were also included in Brennan & Pykkänen’s (2010) models.¹³ In all the

¹³The models in Brennan & Pykkänen’s (2010) Experiment 1 included length of the sentence-initial modifier (in words) as an additional covariate. We pre-registered this as a covariate too;

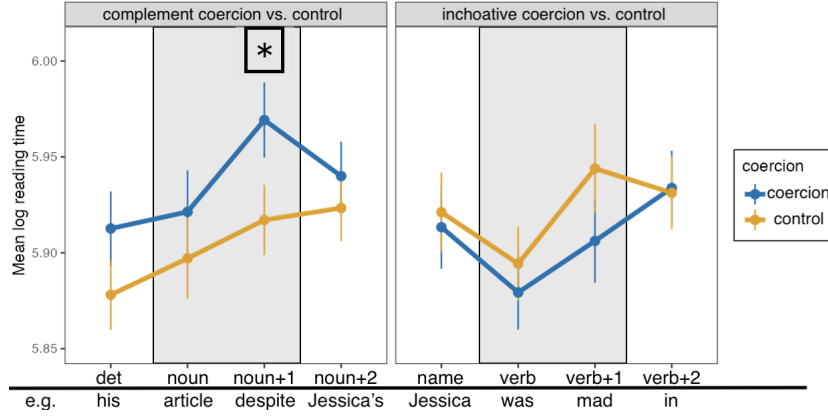


Figure 3: Mean logRTs in Experiment 1a. The regions of interest in each panel are highlighted in grey. Error bars are ± 1 standard error of the mean (SEM). In all of our graphs, asterisks (or lack thereof) reflect the results of our statistical models.

models reported in this paper, we started with the maximal random-effects structure justified by the design (by-participant and by-item random intercepts and slopes for trial type), and reduced the model structure stepwise until we got a converging, non-singular model.¹⁴ The final critical-word model had by-participant and by-item random intercepts; the final critical+1 model had by-participant random slopes for trial type as well. Since we probed for the same effect(s) in two different regions, we used a Bonferroni-corrected significance threshold of $0.05/2 = 0.025$.¹⁵

Using the *faux* package (DeBruine 2025), we coded the “trial type” variable so as to extract three contrasts: (I) inchoative-coercion vs. inchoative-control; (II) complement-coercion vs. complement-control; (III) inchoative-coercion vs. complement-coercion. We are only interested in contrasts (I) and (II), but we had to include a third contrast in the model because the “trial type” variable had four levels. We only interpreted the p-values corresponding to contrasts (I) and (II).

In the model on critical word logRTs, contrasts (I) and (II) were not significant. The model on critical+1 logRTs, by contrast, found that complement-coercion sentences had significantly longer logRTs than matched controls at noun+1 ($\beta = 0.038$,

however, we decided that this was not a sensible predictor to include because half of trials (complement coercion trials and matched controls) did not have a sentence-initial modifier at all, and the other half of trials (inchoative coercion and controls) had modifier lengths of 3-5. In other words, modifier-phrase-length is highly correlated with trial type, making it redundant (multicollinearity).

¹⁴Our pre-registration stated that we would use by-participant and by-item random intercepts only, but it is standard practice to start with the maximal random-effects structure and reduce stepwise (which the lead author did not know at the time); hence, we ultimately opted for this stepwise-reduction procedure.

¹⁵The R pseudo-code for, e.g., the critical+1 model is:
 $\text{logRT} \sim \text{trialtype} + \text{wordLength} + \text{prevlogRT} + \text{freq} + (1 + \text{trialtype} \mid \text{participant}) + (1 \mid \text{item})$.

SE = 0.015, $p = 0.01$), while inchoative-coercion trials’ logRTs at verb+1 did not differ from those of matched controls ($\beta = -0.023$, SE = 0.01, $p = 0.12$).

Figure 4 shows mean naturalness ratings for the four critical trial types. For reference, the mean rating for ungrammatical fillers was 1.72.

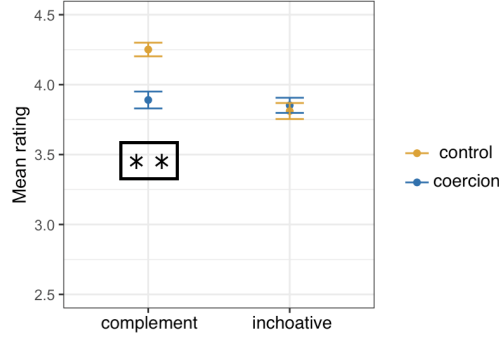


Figure 4: Mean ratings in Experiment 1a. Error bars are ± 1 SEM.

To analyze the naturalness data, we fit a pair of mixed-effects ordinal logistic regression models using the `clmm()` function from the *ordinal* package (Christensen 2023). The models predicted rating from trial type, with by-participant random intercepts and slopes and by-item random intercepts.¹⁶ The first model used data from inchoative coercion trials, matched controls, and ungrammatical fillers; the second used data from complement coercion trials, matched controls, and ungrammatical fillers. Both models used a Helmert coding scheme for the “trial type” variable that allowed us to extract the following contrasts: ungrammatical vs. grammatical (where “grammatical” pools coercion and control) and coercion vs. control. For example, our first model compared (a) inchoative-coercion trials and controls vs. ungrammatical fillers; (b) inchoative-coercion vs. inchoative-control.

Our first model found that ungrammatical fillers were less natural than inchoative coercion trials+controls pooled together ($\beta = -1.77$, SE = 0.32, $p < 0.0001$), but ratings for inchoative-coercion trials did not significantly differ from inchoative-control trials ($\beta = -0.06$, SE = 0.08, $p = 0.4$). Our second model found that ungrammatical fillers were less natural than complement coercion trials+controls pooled ($\beta = -1.88$, SE = 0.33, $p < 0.0001$), and complement coercion trials were less natural than controls ($\beta = -0.48$, SE = 0.09, $p < 0.001$).

4.2.2 Experiment 1b

Figure 5 summarizes RT results for the analyzed subset of Experiment 1b’s data. The left panel displays mean logRTs for complement coercion trials like (c) in Table

¹⁶i.e. rating $\sim$ trialtype + (1 + trialtype | participant) + (1 | item).

2 and controls around the region of interest for complement coercion; the right panel displays logRTs for completive coercion trials like (a) in Table 2 and controls like (b) around the region of interest for completive coercion. Recall that Rett’s theory expects longer logRTs for trials like (a) vis-à-vis trials like (b) at verb or verb+1.

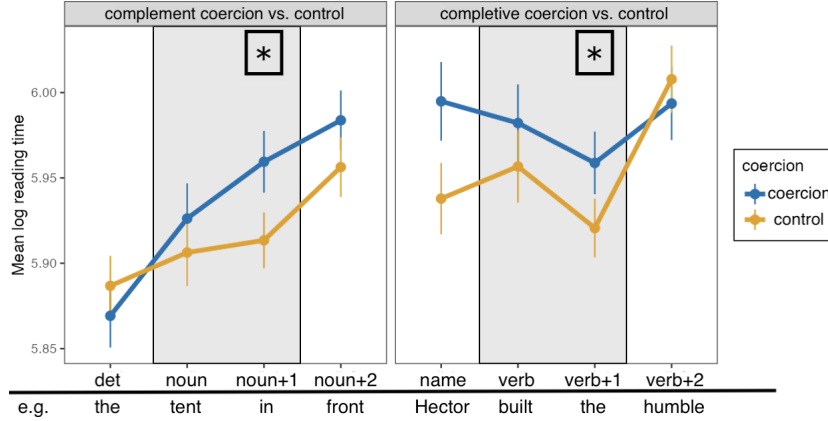


Figure 5: Mean logRTs in Experiment 1b. The regions of interest in each panel are highlighted in grey. Error bars are ± 1 SEM.

We analyzed the RT data using a pair of mixed-effects models (one fit on critical word logRTs, the other fit on critical+1 logRTs). The models had the same fixed-effects structure as the RT models in Experiment 1a; the random-effects structure was determined using the same procedure, with the final structure for both models being by-participant and by-item random intercepts. As in Experiment 1a, we adopted a Bonferroni-corrected significance threshold of $\alpha = 0.025$ and custom-coded the trial type variable so as to extract the following three contrasts: (I) completive-coercion vs. completive-control; (II) complement-coercion vs. complement-control; (III) completive-coercion vs. complement-coercion.

The critical-word model found no significant differences between trial types. The critical+1 model, by contrast, found that completive coercion trials had significantly longer logRTs than matched controls at verb+1 ($\beta = 0.033$, $SE = 0.013$, $p = 0.01$) and that complement coercion trials had significantly longer logRTs than matched controls at noun+1 ($\beta = 0.035$, $SE = 0.013$, $p = 0.01$).

Figure 6 shows mean naturalness ratings for the four critical trial types; for reference, the mean rating for ungrammatical fillers was 1.83.

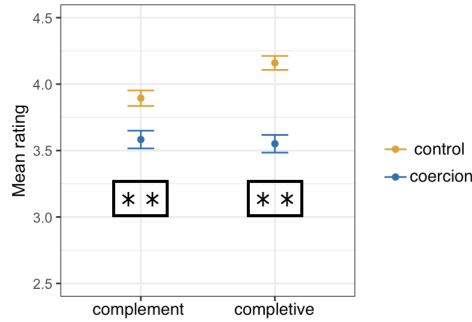


Figure 6: Mean ratings in Experiment 1b. Error bars are ± 1 SEM.

To analyze the naturalness data, we fit two mixed-effects ordinal logistic regression models in the same manner as Experiment 1a. The first model, which used data from completive coercion trials, matched controls, and ungrammatical fillers, found that ungrammatical fillers were less natural than completive coercion + controls pooled ($\beta = -1.4$, $SE = 0.15$, $p < 0.0001$), and completive coercion trials were less natural than controls ($\beta = -0.53$, $SE = 0.08$, $p < 0.0001$). Our other model, which used data from complement coercion trials, matched controls, and ungrammatical fillers, found that ungrammatical fillers were less natural than complement coercion + controls pooled ($\beta = -1.29$, $SE = 0.19$, $p < 0.0001$), and complement-coercion trials were less natural than controls ($\beta = -0.25$, $SE = 0.07$, $p < 0.001$).

4.3 Discussion

Experiment 1a failed to replicate Brennan & Pykkänen’s (2010) findings about atelic sentences with *within*-modifiers: while the atelic *within*-modifier sentences in Brennan & Pykkänen’s (2010) stimuli had slower reading times than controls in the region where the inchoative coercion account expects a slowdown, the atelic *within*-modifier sentences in Experiment 1a did not. This replication failure calls into question the claim that INCHOAT is needed to account for atelic *within*-modifier sentences. Experiment 1a also did not replicate Brennan & Pykkänen’s (2010) finding that atelic *within*-modifier sentences have lower offline ratings than controls. However, despite not finding an inchoative coercion effect, Experiment 1a replicated past work on complement coercion, suggesting that our null result on inchoative coercion cannot be chalked up to an overall inability to capture coercion effects using our paradigm and contexts.

While Experiment 1a did not find evidence for INCHOAT, Experiment 1b did find evidence for COMPLET: sentences with *at*-modifiers + accomplishments had slower reading times than controls just after the main verb. This result lends credence to the claim that COMPLET is needed outside of *before*-clauses. Experiment

1b also replicated past findings about complement coercion, which serves as an additional sanity check. Finally, Experiment 1b found that sentences with completive and complement coercion, like certain other cases of coercion (see section 2.2), are slightly less natural than controls.

At this point, the major question that remains about Experiment 1 concerns why the results of Experiment 1a conflicted with those of Brennan & Pykkänen (2010). We can think of two possibilities. First, while Experiment 1a only compared *within*-adverbial sentences to minimally different *for*-adverbial sentences, many of Brennan & Pykkänen’s (2010) comparisons were less minimal (see section 3.1). Brennan & Pykkänen (2010) report some unexpected differences between the processing of their inchoative coercion sentences and the processing of their controls; for instance, their controls were harder to process than their inchoative coercion sentences just before the critical region. Hence, it is possible that the use of non-minimal comparisons in Brennan & Pykkänen (2010) may underlie the differences in our results. Second, Experiment 1a’s target sentences used *be*-statives, whereas Brennan & Pykkänen’s used subject experiencer verbs. The INCHOAT-based theory does not predict a processing asymmetry between these kinds of sentences, but it is nonetheless conceivable that self-paced reading is only sensitive enough to detect inchoative coercion’s effect when it occurs in subject-experiencer sentences.

In the remainder of this paper, we follow up on the results of Experiments 1b and 1a in turn.

5 Experiment 2

Since Experiment 1b detected a completive coercion effect in *at*-modifier sentences in self-paced reading, Rett (2020) would expect that completive coercion in *before*-clauses would have a similar self-paced reading cost. Experiment 2 tests this prediction, looking at the processing of accomplishments in *before*-clauses in two types of contexts: (a) contexts that are only compatible with a before-finish reading (i.e. contexts that, according to Rett (2020), require coercion); (b) contexts that are compatible with a before-start reading and thus should not require coercion.

5.1 Methodology

5.1.1 Participants and materials

120 self-reported native English speakers from Prolific participated in Experiment 2, each of whom was paid £4.50 for their participation regardless of whether they were ultimately included in the analyzed sample. Abnormally long and short RTs (defined as in Experiment 1, see section 4.2) were excluded from analysis, as well

as data from participants who gave a “True” rating on at least three of the six false fillers. 30 participants were excluded under the latter criterion.

The structure of each trial in Experiment 2 was the same as in Experiment 1 (context, then target sentence, then rating). However, Experiment 2 had three critical trial types rather than four (see **Table 3**). On no-coercion trials like (a) in Table 3, participants read target sentences with accomplishments in *before*-clauses after seeing before-start contexts like Context A, in which Emma’s irritation emerged before Hector started building the tent. Completive coercion trials like (b) used the same target sentences as no-coercion trials but presented them alongside before-finish contexts like Context B, in which Emma’s irritation emerged after Hector started building the tent but before he finished. Finally, complement coercion trials like (c) presented complement-coercion target sentences with *finish* after contexts like Context B. As in Experiment 1, complement coercion trials were included as a sanity check; following past literature, we expect longer RTs for (c)’s target sentence vis-à-vis (a)’s target sentence at or just after *tent*.

- (29) a. **Context A:** Hector is a lifelong hiker and his daughter Emma wants to learn more about the outdoors, so they went on a hiking trip in the woods. It got very windy, so they decided to build a tent and wait in it until the wind stopped. Hector pulled out the camping equipment at 4pm and took longer than usual to set the tent up because of the wind; however, the tent was fully up by 4:30pm. Emma, who had become very irritable when the wind started (2pm), regained her composure once she saw the full tent at 4:30pm.
- b. **Context B:** Same first three sentences as Context A, then: Emma, who had become very irritable when Hector made a big mistake and had to start over at 4:15pm, regained her composure once she saw the full tent at 4:30pm.

Trial Type	Context	Target Sentence
(a) no-coercion	A	Emma was irritable before Hector built the tent in the woods.
(b) completive	B	Emma was irritable before Hector built the tent in the woods.
(c) complement	B	Emma was irritable before Hector finished the tent in the woods.

Table 3: Sample stimuli for Experiment 2. A sample trial is available at [this link](#).

As participants were reading a context, they saw timelines to help them keep track of the events that would be mentioned in the upcoming target sentence (see Figure 7). For example, on Emma-and-Hector trials (Table 3), a timeline showing the duration of Hector building the tent appeared once participants started to read the third sentence of Context A or B (Figure 7i). Once they moved on to

the final sentence of the context (i.e. the sentence about the duration of Emma’s irritation), the timeline morphed into either the one in Figure 7ii (if they were reading Context A) or the one in Figure 7iii (for Context B). These timelines were included because we felt that it would be hard to keep track of the information in the context otherwise.

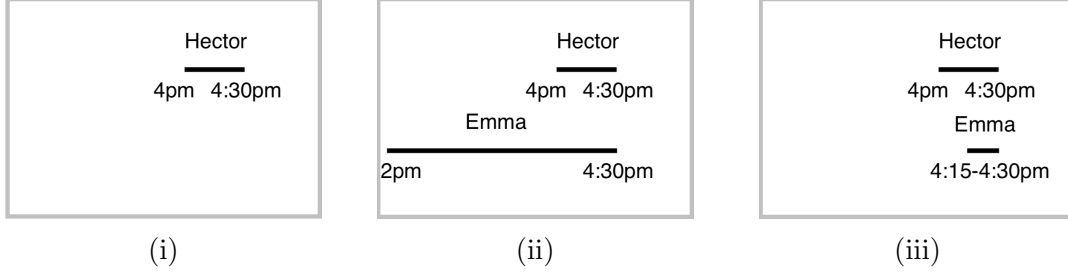


Figure 7: Timelines used on the trials in Table 3.

As in Experiment 1, there were both ungrammatical and grammatical fillers, some of which were true given the context and others of which were not.

One further difference between Experiment 2 and Experiment 1 bears mention. In Experiment 1, participants rated the naturalness of the target sentence on every trial they saw. In Experiment 2, participants rated the naturalness of the target sentence on critical trials and ungrammatical fillers but had to make a “True or False” judgment instead on grammatical fillers. We made this choice because we hypothesized that “True or False” judgments might be a better way of gauging participants’ attention than Likert ratings (see fn. 9).¹⁷

5.1.2 Procedure

After providing consent, participants saw an instruction text and a practice trial (accessible [here](#)). Then, participants were sorted into one of four lists, each of which contained 12 critical trials (4 no-coercion trials, 4 completive coercion trials, 4 complement coercion trials) and 12 fillers (2 ungrammatical fillers, 4 grammatical+true fillers, and 6 grammatical+false fillers). The ungrammatical fillers and grammatical+true fillers were the same across the four lists, as were two of the grammatical+false fillers. The remaining four grammatical+false fillers in each list drew from the same pool of contexts as critical trials but used false *after*-sentences (e.g.

¹⁷In spite of this hunch, the exclusion rate (30/120, see the first paragraph of this section for details) was in the same ballpark as Experiment 1’s, which suggests that Experiment 1’s high exclusion rate was not due to its use of Likert-item attention checks but rather due to, e.g., participant fatigue. Whatever the cause of these high exclusion rates, we believe the results suggest that our manipulations were effective for attentive participants. In Experiment 3, we returned to Likert-item exclusions but used a slightly more lenient exclusion criterion.

Context A in Table 3 + *Emma was irritable after Hector built the tent in the woods*). For these fillers and the critical trials, we created 16 item sets like the item set in Table 3 (each centered around a different accomplishment) and used the Latin-square pattern detailed in section 4.1.2 to ensure that each participant saw one trial from each item set. For example, participants in list A saw an Emma-and-Hector context paired with a false *after*-sentence, participants in list B saw a completive-coercion Emma-and-Hector trial, participants in list C saw a no-coercion Emma-and-Hector trial, and participants in list D saw a complement-coercion Emma-and-Hector trial.

5.2 Results

Figure 8 summarizes RT results for complement coercion trials (e.g. (c) in Table 3) vs. no-coercion trials like (a) in and around the critical region for complement coercion (direct object noun and noun+1 in the *before*-clause).

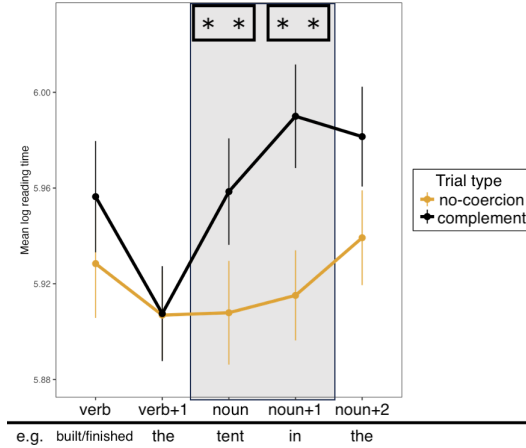


Figure 8: Mean logRTs in Experiment 2, no-coercion vs. complement. Error bars are ± 1 SEM.

To ascertain whether Experiment 2 replicated past findings on complement coercion, we fit two mixed-effects models on the data from complement and no-coercion trials. The first model was fit on logRTs at the direct object noun, while the second examined logRTs at noun+1. The models predicted logRT from trial type (complement coercion vs. no-coercion), with the same covariates as Experiment 1 and the random-effects structure being determined using the same procedure; see Supplementary Materials for details. Since we probed for a complement coercion effect in two separate regions, we adopted a Bonferroni-corrected significance threshold of $\alpha = 0.025$ for these models. Complement trials had significantly longer logRTs than no-coercion trials at both noun ($\beta = 0.058$, $SE = 0.019$, $p = 0.002$) and noun+1 ($\beta = 0.057$, $SE = 0.017$, $p = 0.001$).

Figure 9 displays logRTs for complete trials (e.g. (b) in Table 3) and no-coercion trials in the *before*-clause. Recall that under Rett’s (2020) theory, we predict a slowdown for complete trials relative to no-coercion, and we *prima facie* expect this slowdown to occur at either the main verb in the *before*-clause or verb+1.

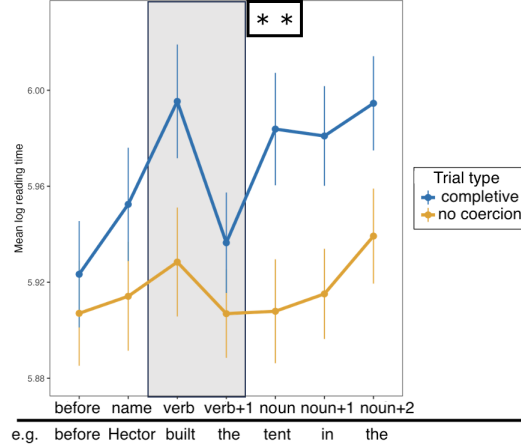


Figure 9: Mean logRTs in Experiment 2, no-coercion vs. complete. Error bars are SEM.

For each of the critical words for complete coercion (verb, verb+1), we fit mixed models with the same structure as above on the data from complete and no-coercion trials, finding no significant logRT differences between the two trial types at verb ($\beta = 0.036$, $SE = 0.022$, $p = 0.11$) or verb+1 ($\beta = 0.008$, $SE = 0.018$, $p = 0.67$). Note that although the difference between no-coercion vs. complete at verb looks sizable in Figure 9, the verb model is telling us that this effect is not robust when controlling for factors like logRT on the previous word.

In addition to the above analyses, which tested our pre-registered hypotheses, we performed an exploratory analysis. In particular, since visual inspection of Figure 4 suggests a sizable slowdown for complete coercion trials (relative to no-coercion) at verb+2 (aka “noun”) and onward, we fit another model on verb+2 RTs for complete and no-coercion trials (same structure as above). The model found that complete trials were read significantly more slowly than no-coercion trials at verb+2 ($\beta = 0.06$, $SE = 0.02$, $p = 0.003$). Note that this effect is still significant even when we Bonferroni-correct for the fact that we probed for a complete coercion effect in three different regions ($\alpha = 0.05/3 = 0.016$).¹⁸

¹⁸As an additional exploratory analysis, we checked to see if there was within-experiment learning that led the complete coercion effect to become delayed or less localized as the experiment went on. Our main finding was that if we fit a model on the subset of the data corresponding to the first no-coercion trial and complete trial that each participant saw, there is a marginally

Finally, Figure 10 shows mean naturalness ratings for the three critical trial types. For reference, the mean rating for ungrammatical fillers was 1.62.

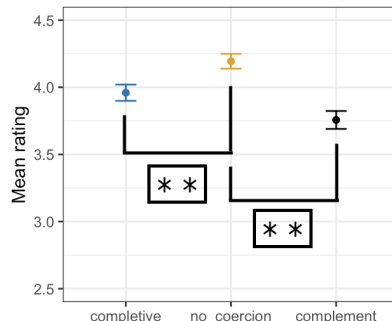


Figure 10: Mean naturalness ratings, Experiment 2. Error bars are ± 1 SEM.

To analyze the naturalness data, we fit a mixed-effects ordinal logistic regression on the three critical trial types. The model predicted rating from trial type (no-coercion as reference level), with by-participant and by-item random intercepts and random slopes for trial type. The model found that completeive trials received significantly lower ratings than no-coercion trials ($\beta = -0.72$, $SE = 0.23$, $p = 0.001$), and complement trials also received significantly lower ratings than no-coercion trials ($\beta = -1$, $SE = 0.25$, $p < 0.0001$).

5.3 Discussion

Experiment 2 straightforwardly replicated past findings on complement coercion, verifying that we can capture coercion effects using our paradigm and stimuli. Additionally, while Rett’s (2020) predicted slowdown in completeive trials did occur, it started later than expected: past studies on aspectual coercion have found processing slowdowns at the verb or verb+1, whereas the slowdown for completeive trials started at verb+2.

We can think of two ways for a proponent of Rett’s (2020) theory to explain why the completeive coercion effect in Experiment 2 was delayed to verb+2 (and why the complement coercion effect was not similarly delayed); the viability of either of these explanations needs to be assessed further, however. The first explanation concerns certain differences between the $n+1$ spillover word for the completeive coercion effect vis-à-vis the $n+1$ spillover word for complement coercion. The $n+1$ spillover word for completeive coercion, i.e. verb+1, was always *the*. Not only is

significant slowdown for completeive trials at verb ($\beta = .101$, $SE = 0.05$, $p = 0.045$). Although $p < 0.05$ in this model, we opt not treat this effect as significant because the p-value is above a Bonferroni-corrected significance threshold.

this word short and maximally frequent, but it was fully predictable in our stimuli: fillers and critical trials alike all had *the* in this position. Participants may have internalized the predictable nature of verb+1 as the experiment went on; as a result of the predictability, frequency, and short length of our verb+1, perhaps participants breezed through this word on many trials before the processing difficulty engendered by the verb had time to manifest behaviorally. If this explanation is right, any spillover effects from the verb should show up on verb+2 rather than verb+1.¹⁹

Although the $n+1$ spillover word for the complement coercion effect, i.e. noun+1, was always short and high-frequency as well, it was much less predictable: the preposition or complementizer used as noun+1 was not the same across items. Thus, participants had less of an incentive to breeze through noun+1 vis-à-vis verb+1. In sum, the idea is that for completive coercion, any spillover effects due to shallow parsing on later trials are likely to show up on verb+2 rather than verb+1, as verb+1 is too light and predictable to soak up any processing load; however, a parser who skims over verb+1 might not skim over noun+1 (since noun+1 is less predictable), meaning that any spillover effects for complement coercion are more likely to appear on noun+1.

While this is one type of explanation for the completive vs. complement asymmetry, it is not the only one. Another, less deflationary type of explanation relies on the fact that, as mentioned in section 3, completive coercion in *before*-clauses must be thought of as pragmatic (i.e. needed to repair a mismatch between a sentence’s truth-conditions and the context), whereas complement coercion must be thought of as semantic (i.e. needed to repair a sentence-internal aspectual mismatch). Perhaps the cost incurred by “pragmatic” coercions (whose processing has not been studied before now) is slightly delayed relative to “semantic” coercions because it takes longer to realize that there is a mismatch between a sentence and a context than it does to realize that there is a sentence-internal aspectual mismatch. On this view, the costs observed in Experiment 1b and 2 both reflect retroactive insertion of a coercion operator, but the cost of COMPLET-insertion in *before*-clauses is delayed relative to *at*-modifier sentences because it takes longer to realize that operator insertion is necessary in the former than in the latter. This explanation dovetails with certain psycholinguistic theories, which posit a lag between semantic and pragmatic processing (Huang & Snedeker 2009, a.o.).

To sum up, Experiment 2 suggests that vis-à-vis before-start readings of telic *before*-clauses, before-finish readings incur a cost both in online processing and in offline ratings, though the cost was delayed relative to expectations and thus needs to be investigated further. Rett (2020) is the only theory of *before* that predicts

¹⁹Note that verb+1 was slightly less predictable in Experiment 1 (almost half of fillers had determiners other than *the*), which may explain why participants did not breeze through verb+1 in Experiment 1.

any asymmetry between the two readings (see General Discussion), and hence Rett (2020) has a leg up on competitors when it comes to explaining the results of Experiment 2, even though some questions remain open for her as well.

6 Experiment 3

Having tested the processing predictions that Rett’s (2020) theory makes about completive coercion, we now follow up on Experiment 1a’s results on inchoative coercion. Recall that Experiment 1a failed to replicate Brennan & Pylkkänen’s (2010) finding that sentences with atelic predicates and *within*-modifiers incur a processing cost, and that this replication failure could have multiple sources, including (a) the use of non-minimal comparisons in Brennan & Pylkkänen (2010); (b) Experiment 1a’s use of *be*-statives rather than subject-experiencer verbs.

To assess whether our use of *be*-statives underlies our replication failure vis-à-vis Brennan & Pylkkänen (2010), we decided to repeat Experiment 1a but use subject-experiencer verbs. If we continue to find an online effect for complement coercion and find no comparable inchoative coercion effect despite using Brennan & Pylkkänen’s (2010) predicates, our null results on inchoative coercion cannot be chalked up to our predicate choice, and there would no longer be clear processing evidence for an INCHOAT-based theory of atelic *within*-modifier sentences.

6.1 Methodology

As in Experiment 1a, we recruited 110 self-reported native English speakers from Prolific, each of whom was paid £4.50 for their participation. Abnormally long and short RTs (defined as above) were excluded, along with data from the 14 participants who gave a rating of 3 or higher on at least 4 out of 6 false/ungrammatical fillers.

The design and procedure of Experiment 3 were identical to Experiment 1a in all respects except for the fact that Experiment 3 used subject-experiencer sentences where Experiment 1a used *be*-statives. This change necessitated some slight alterations to the contexts, which were kept as minimal as possible. As an example of Experiment 3’s stimuli, consider the item set in **Table 4**, which is Experiment 3’s analogue of Table 1.

- (30) a. **Context A:** One semester, Jessica took a class taught by a professor, Alex, whose lectures were extremely difficult to understand. Jessica had heard that Alex was more coherent in office hours, so she scheduled a 20 minute appointment with him. Alex did nothing but read one of his own research articles to Jessica in full; Jessica voiced her annoyance about Alex’s behavior at first, but by the fifteen-minute mark, Jessica had given up and decided to tolerate Alex.

- b. **Context B:** Same first two sentences as Context A, then: Alex did nothing but read one of his own research articles to Jessica in full; Jessica tolerated Alex during the first fifteen minutes, but after that, she snapped and voiced her annoyance—all while Alex kept on reading.

Trial type		Context	Target sentence
Inchoative	(a) Coercion	A	Within fifteen minutes Jessica tolerated the unhelpful professor.
	(b) Control	B	For fifteen minutes Jessica tolerated the unhelpful professor.
Complement	(c) Coercion	A/B	Alex continued his article despite Jessica’s protest.
	(d) Control	A/B	Alex read his article despite Jessica’s protest.

Table 4: Sample stimuli for Experiment 3. A sample trial is available [here](#).

6.2 Results

Figure 11 summarizes RT results for the analyzed subset of the data, both for complement-coercion trials vs. matched controls (left panel) and inchoative-coercion vs. matched controls (right panel).

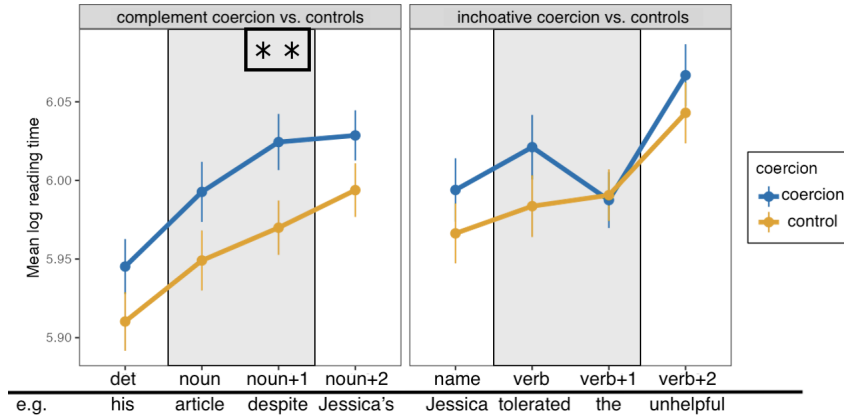


Figure 11: Mean logRTs in Experiment 3. Error bars are ± 1 SEM.

Recall that under the inchoative coercion theory of *within*-modifier sentences, we expect longer RTs for inchoative coercion trials relative to controls at either the verb (e.g. *tolerated*) or verb+1. To ascertain whether this prediction was borne out (and to assess whether Experiment 3 replicated past findings on complement coercion), we analyzed the RT data in the same way as in Experiment 1a (see section 4.2), fitting one model on critical word logRTs and one model on critical+1 logRTs. In the model on critical word logRTs, the difference between inchoative coercion

sentences and matched controls was not significant ($\beta = 0.02$, $SE = 0.015$, $p = 0.17$), and neither was the difference between complement coercion sentences and matched controls ($\beta = 0.025$, $SE = 0.015$, $p = 0.11$). In the model on critical+1 logRTs, by contrast, the difference between complement coercion sentences and matched controls was significant ($\beta = 0.035$, $SE = 0.013$, $p = 0.009$), though the difference between inchoative coercion sentences and matched controls was still insignificant ($\beta = -0.003$, $SE = 0.013$, $p = 0.78$).

Figure 12 shows mean naturalness ratings for the four critical trial types. For reference, the mean naturalness rating for the ungrammatical fillers was 1.58.

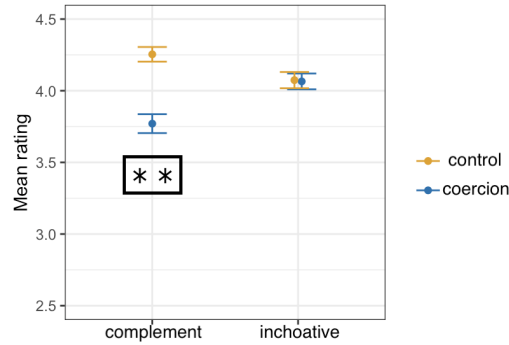


Figure 12: Mean naturalness ratings in Experiment 3. Error bars are ± 1 SEM.

We analyzed the naturalness data as in Experiment 1a. Our first model found that while ungrammatical fillers were less natural than inchoative coercion trials + controls pooled together ($\beta = -1.8$, $SE = 0.26$, $p < 0.0001$), there was no significant difference in ratings for inchoative coercion sentences vs. controls ($\beta = -0.035$, $SE = 0.07$, $p = 0.63$). Our second model found that ungrammatical fillers were less natural than complement coercion trials + controls pooled together ($\beta = -0.6$, $SE = 0.1$, $p < 0.0001$), and complement coercion trials were less natural than matched controls ($\beta = -1.74$, $SE = 0.3$, $p < 0.0001$).

6.3 Discussion

Despite using the same types of predicates as Brennan & Pylkkänen (2010), Experiment 3—like Experiment 1a—failed to replicate Brennan & Pylkkänen’s findings about atelic *within*-modifier sentences. This suggests that the failure to replicate Brennan & Pylkkänen (2010) in Experiment 1a was not due to our predicate choice but some other factor, e.g. the use of non-minimal comparisons in Brennan & Pylkkänen (2010) (see section 4.3). Despite not finding a cost for inchoative coercion, Experiment 3 found an effect of complement coercion, suggesting that the

results on inchoative coercion in Experiment 3 cannot be due to an overall inability to capture coercion effects with our contexts.

Whatever its source turns out to be, this continued failure to replicate Brennan & Pytkäinen (2010) calls a coercion-based theory of atelic *within*-modifier sentences into question. See section 8.2 for discussion of an alternative, non-coercive theory of these sentences.

7 Experiment 4

Regardless of the ultimate truth about atelic *within*-modifier sentences, it is still possible that there is coercion in atelic *after*-clauses where Rett (2020) claims there is. In other words, even if there turns out to be no inchoative coercion in atelic *within*-modifier sentences, we can still test the idea that the after-start reading of atelic *after*-clauses has a coercion operator that the after-finish reading lacks. Experiment 4 tests this idea.

7.1 Methodology

7.1.1 Participants and materials

125 self-reported native English speakers from Prolific participated in Experiment 4, each of whom was paid £4.50 for their participation. Abnormally long or short RTs (defined as in section 4.2) were excluded from analysis, as were data from participants who gave a rating of 3 or higher on at least four out of the eight ungrammatical/false fillers. 40 participants were excluded under this criterion.

The structure of each trial in Experiment 4 was the same as in past experiments (context, then target sentence, then rating). Experiment 4 had four critical trial types (see **Table 5**).²⁰ On inchoative coercion trials like (a) in Table 4, participants read target sentences with subject-experiencer predicates in *after*-clauses alongside after-start contexts like Context A, in which Dave was regretful after Lara started fearing the dog but not after she stopped. Inchoative controls like (b) in Table 5 used the same target sentences as inchoative coercion trials but presented them alongside after-finish contexts like Context B, in which Dave was regretful even after Lara stopped fearing the dog. Unlike in Experiment 2, it was not feasible to implement a complement coercion sanity check by modifying the target sentences used in inchoative trials, as one cannot replace *after Lara feared the dog* with, e.g., *after Lara started the dog* and preserve the subject-experiencer event sense. As an

²⁰Note that the items in Experiment 4 were similar to those in Experiments 1a and 3. Thus, the Dave-and-Lara item in Table 5 had an analogue in Experiments 1a and 3, and the Alex-and-Jessica item in Tables 1 and 4 had analogues in Experiment 4. We show the Dave-and-Lara item here for a bit of variation.

alternative, we implemented a complement coercion sanity check by comparing the processing of, e.g., (c) and (d) in Table 4 in slightly different contexts like Context C. All complement coercion trials and matched controls used contexts without overlap between the matrix and embedded eventualities.

- (31) a. **Context A:** Lara, a physicist, was recently invited to present her work to Dave, a famous physicist and Nobel Prize winner, in a conference hall. Dave ended up bringing a large dog, and Lara is afraid of dogs. While Lara felt fear towards the dog during the first fifteen minutes of her presentation (10-10:15am), she calmed down at the 15-minute mark and managed to immerse herself in her ideas. Dave had started to feel regretful about bringing the dog at 10:05, when he noticed how afraid Lara was; those feelings of regret went away once Lara regained her composure at 10:15.
- b. **Context B:** Same first three sentences as Context A, then: Dave had started to feel regretful about bringing the dog at 10:05, when he noticed how afraid Lara was; those feelings of regret lasted for many days, as he was sorry for having given Lara a few minutes of fear.
- c. **Context C:** Same first two sentences as Context A, then: So she smoked a cigarette outside the building (9:50-9:55am) to calm her nerves before the presentation. Dave passed by Lara at 9:56 and smelled the smoke; Dave hates smokers, so he immediately felt regretful about agreeing to meet Lara.

Trial type		Context	Target sentence
Inchoative	(a) Coercion	A	Dave was regretful after Lara feared the large dog.
	(b) Control	B	Dave was regretful after Lara feared the large dog.
Complement	(c) Coercion	C	Dave was regretful after Lara enjoyed a cigarette outside the building.
	(d) Control	C	Dave was regretful after Lara smoked a cigarette outside the building.

Table 5: Sample stimuli for Experiment 4. Click [here](#) to run through these four trials yourself.

A few further notes about the design bear mention. First, we opted to use subject-experiencer verbs in Experiment 4 rather than *be*-statives simply because these were the predicates used by Brennan & Pylkkänen (2010) in their work on inchoative coercion; Experiments 1a and 3 suggest that this choice is likely not consequential. Second, as in Experiment 2, participants saw timelines as they were reading a context to help them keep track of the events that would be mentioned in the upcoming target sentence. For example, on trials (a) and (b) in Table 5,

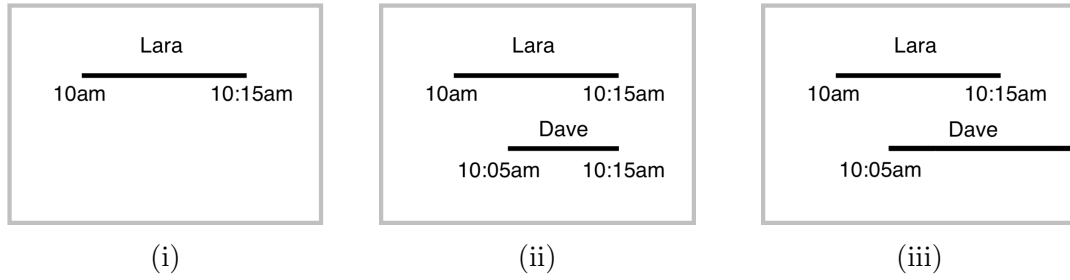


Figure 13: Timelines used on trials (a) and (b) in Table 4. Trials (c) and (d) used similar timelines for Dave’s regretfulness and Lara’s smoking.

participants would see a timeline showing the duration of Lara’s fear (Figure 13i) once they started to read the third sentence of Context A or B. Once they moved on to the final sentence (i.e. the sentence about the duration of Dave’s regretfulness), the timeline morphed into either the one in Figure 13ii (if they were reading Context A) or 13iii (if they were reading Context B).

As in past experiments, there were both ungrammatical fillers with selectional violations and grammatical fillers, some of which used target sentences that correctly described the temporal order of events in the scenario and some of which did not.

7.1.2 Procedure

After providing consent, participants saw the same instruction text and practice trial as Experiment 2. Upon completing the practice trial, participants were sorted into one of four lists, each of which contained 16 critical trials (4 x 4 types) and 10 fillers (2 ungrammatical, 6 grammatical and false, 2 grammatical and true). The fillers were the same across the four lists; for the critical trials, we created 16 item sets like the one in Table 5 and used a Latin-square pattern to ensure that each participant saw one trial from each item set.

7.2 Results and analysis

Figure 14 summarizes RT results for complement coercion trials (e.g. (c) in Table 5) and matched controls like (d) in and around the region of interest for complement coercion. Descriptively, we see a slowdown for complement coercion trials, but it appears to start earlier than expected (at “det”). To analyze these results, we fit two models on the data from complement and no-coercion trials: one on direct object noun logRTs, and one on noun+1 logRTs. The models predicted logRT from trial type (complement-coercion vs. complement-control), with the same covariates as in past experiments and the random-effects structure being determined using the same procedure. As in past experiments, we adopted a significance threshold of $\alpha = 0.025$

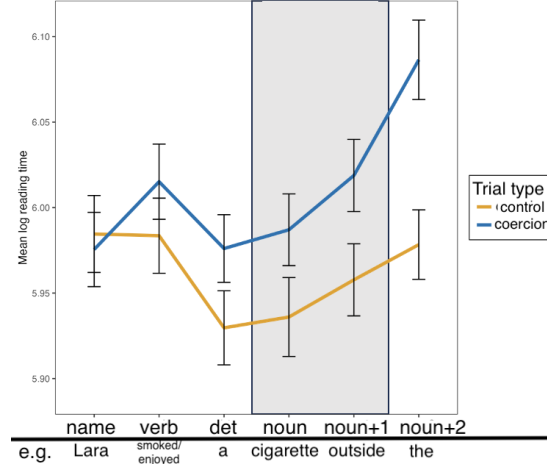


Figure 14: Mean logRTs in Experiment 4, complement coercion vs. matched controls.

in these models to correct for the fact that we probed for a complement coercion effect in two regions. The models did not find a significant effect at noun ($\beta = 0.02$, $SE = 0.017$, $p = 0.24$) or noun+1 ($\beta = 0.02$, $SE = 0.023$, $p = 0.35$). Note that since previous word logRT is being included as a covariate, these non-significant results should be interpreted as suggesting that the complement coercion effect at noun and noun+1 is not significantly greater than the effect at the determiner; this dovetails with what we see in Figure 14.

Figure 15 displays logRTs for inchoative coercion trials (e.g. (a) in Table 5) and matched controls like (b) in the *after*-clause.

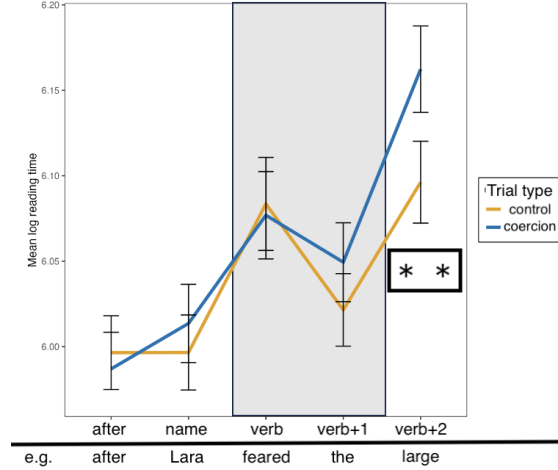


Figure 15: Mean logRTs in Experiment 4, inchoative coercion vs. matched controls.

For each of the *a priori* critical regions for inchoative coercion, we fit a mixed-

effects model with the same structure as above on the data from inchoative coercion trials and matched controls. The models did not find a significant difference between the two trial types at verb ($\beta = -0.014$, $SE = 0.026$, $p = 0.58$) or verb+1 ($\beta = 0.03$, $SE = 0.019$, $p = 0.11$).

In addition to the above analyses, which tested the hypotheses that were developed prior to any experimentation (see section 3), we performed the same sort of exploratory analysis as in Experiment 2, fitting a model on verb+2 logRTs for inchoative coercion trials vs. matched controls (same model structure as above). The model found that inchoative coercion trials were read significantly more slowly than matched controls at verb+2 ($\beta = 0.066$, $SE = 0.024$, $p = 0.008$). Note that this effect is still significant even when we Bonferroni-correct for the fact that we probed for an inchoative coercion effect in three regions ($\alpha = 0.05/3 = 0.016$).²¹

Finally, Figure 16 shows mean naturalness ratings for the four critical trial types. For reference, the mean rating for ungrammatical fillers was 2.27.

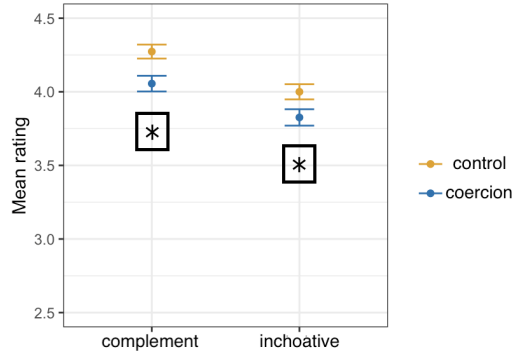


Figure 16: Mean naturalness ratings, Experiment 4.

We analyzed the naturalness data in the same way as in Experiments 1a and 3. Our first model found that ungrammatical fillers were less natural than inchoative coercion trials + controls pooled together ($\beta = -0.96$, $SE = 0.13$, $p < 0.0001$), and inchoative coercion trials were less natural than controls ($\beta = -0.16$, $SE = 0.07$, $p = 0.01$). Our second model found that ungrammatical fillers were less natural than complement coercion trials + controls pooled together ($\beta = -1.09$, $SE = 0.22$, $p < 0.0001$), and complement coercion trials were less natural than controls ($\beta = -0.16$, $SE = 0.075$, $p = 0.03$).

²¹As in Experiment 2, we also performed some exploratory data analysis to see if there were any within-experiment changes in the inchoative coercion effect. Descriptive statistics suggested no strong changes, so we did not fit any more models to avoid inflating the number of comparisons.

7.3 Discussion

Experiment 4’s results on *after*-clauses are akin to Experiment 2’s results on *before*-clauses. For one thing, inchoative coercion trials in Experiment 4 and completive trials in Experiment 2 were both less natural than controls, like some other cases of coercion. Furthermore, in both experiments, Rett’s (2020) predicted slowdown occurred but started later than expected. In particular, the predicted slowdown for inchoative coercion trials in Experiment 4, like the predicted slowdown for completive trials in Experiment 2, occurred at verb+2 rather than at verb or verb+1. The only way in which Experiment 4’s results differ from Experiment 2’s has to do with complement coercion: while Experiment 2 straightforwardly replicated past findings on this type of coercion, the complement coercion effect in Experiment 4 was diffuse and not localized to the noun or noun+1, and hence our pre-registered analyses did not find a significant effect.

As in Experiment 2 (see section 5.3), there are two main ways for a proponent of Rett’s (2020) theory to explain why we saw an inchoative coercion effect at verb+2 in Experiment 4 rather than at verb or verb+1, though the feasibility of these explanations needs to be assessed further. The first explanation concerns the nature of verb+1 in Experiment 4. While verb+1 in Experiment 4 was not always *the* (cf. Experiment 2), it was always *the*, *a*, *his*, or *her*, i.e. a short, frequent, predictable determiner. As such, it is possible that participants breezed through verb+1 on many trials without stopping to process, meaning that any leftover processing load from the verb would spill over onto verb+2 rather than verb+1.

Alternatively, one could interpret the later-than-expected slowdown in inchoative coercion trials as reflecting the fact that inchoative coercion in *after*-clauses is performed to resolve a different kind of mismatch than, e.g., complement coercion (“semantic” vs. “pragmatic” mismatches; see section 4.3 for more on this idea). Regardless of which of these explanations is correct, Experiment 4 suggests that vis-à-vis after-finish readings of atelic *after*-clauses, after-start readings are associated with an online cost and depressed naturalness ratings. Even though the timing of the cost raises additional questions for Rett (2020), she is nonetheless better-positioned than her competitors to explain the results of Experiment 4, as hers is the only existing theory to predict that after-start readings should be more costly than after-finish readings.

Regarding complement coercion, the messy results for this coercion type in Experiment 4 likely reflect an issue with its complement coercion stimuli rather than an issue with the experiment in general. To see this, consider (c)-(d) from Table 5. For a parser, the main verb in (c) from Table 5, i.e. *enjoyed*, is likely much more surprising than the main verb in (d) given the context (Context C), since Dave’s regretfulness was caused not by the fact that Lara enjoyed the cigarette but rather from the fact that she smoked it at all. This difference in the predictability

of the verb, which recurs in many of Experiment 4’s complement-coercion trials and controls, may have triggered a processing slowdown in its own right, leading to the sort of earlier-than-expected effect we see in Figure 14. By contrast, the main verb in Experiment 2’s complement-coercion stimuli, which was always *finished*, is intuitively not much more surprising than the main verb in the no-coercion stimuli; hence, the verb did not trigger a slowdown in that experiment.

8 General discussion

8.1 Main takeaways

This research sought to test a series of processing predictions that Rett’s (2020) theory of *before/after* makes and competing theories do not. The processing predictions in question concern two coercion operators posited by Rett (2020): COMPLET and INCHOAT. For Rett (2020), COMPLET is required in accomplishment *at*-modifier sentences and certain *before*-clauses; INCHOAT is required in atelic *within*-modifier sentences and certain *after*-clauses. Past work has shown that coercion has a distinctive psycholinguistic profile, and as such, Rett (2020) predicts that sentences with COMPLET or INCHOAT should have this profile. Experiments 1b and 2 investigated whether Rett’s (2020) claimed examples of COMPLET-insertion have the psycholinguistic profile of other coercions, and Experiments 1a and 3-4 investigated whether the claimed cases of INCHOAT-insertion do.

Taken together, the results of Experiments 1b and 2 are compatible with Rett’s (2020) theory of completive coercion, though the results of Experiment 2 require her to make some additional assumptions that need to be probed further. Experiment 1b found a processing cost in accomplishment *at*-modifier sentences in the region where the completive coercion account expects a slowdown. This establishes that there is an online cost for completive coercion that is detectable in self-paced reading. Given this finding, Rett (2020) expects that COMPLET-insertion in *before*-clauses should have a self-paced reading cost too. Experiment 2 tested this prediction and found that *before*-clauses that require COMPLET-insertion on Rett’s (2020) theory were indeed harder to process than other *before*-clauses; however, this processing cost started later than expected. Although the timing of the online effect in Experiment 2 was unexpected, there are multiple ways to interpret the results of Experiment 2 under Rett’s theory (see section 5.3 for details). The first Rett-compatible explanation posits that the delay was an experimental artifact, while the other posits that instances of coercion that resolve a pragmatic mismatch have a slightly delayed cost relative to instances of coercion that resolve a semantic mismatch.

Turning to inchoative coercion, the results of Experiments 1a and 3-4 are compatible with some aspects of Rett’s (2020) theory of inchoative coercion but not

others. Experiments 1a and 3 failed to replicate Brennan & Pylkkänen’s (2010) finding that atelic *within*-modifier sentences incur a processing cost. While this replication failure does not rule out the idea that after-start readings of *after*-clauses involve coercion, it does suggest that using atelic *within*-modifier sentences to motivate a coercion-based account of *after*-clauses is likely untenable. Experiment 4 investigates whether there is indeed coercion in *after*-clauses. While *after*-clauses that require INCHOAT-insertion for Rett (2020) were harder to process than other *after*-clauses, this processing difficulty started one word later than expected (much as in Experiment 2). A proponent of Rett (2020) could interpret this delay in either of the two ways that they could interpret the delay in Experiment 2.

Although our results necessitate further assumptions on the part of Rett (2020), they are more compatible with Rett’s (2020) theory than with under-specification theories. Experiments 2 and 4 suggest that there is a processing and rating asymmetry between the two readings of telic *before*-clauses and between the two readings of atelic *after*-clauses; as it stands, under-specification theories do not predict any asymmetries of this sort, since they draw no distinction between the types of *before*-clauses and *after*-clauses that Rett (2020) distinguishes.

Is there a way to modify under-specification theories so that they account for our results? Well, while a proponent of an under-specification theory could try to account for our results by appealing to an implicature, we think that such an approach ultimately does not work. In particular, suppose we try to augment the under-specification theory from section 2.1.1 with the further assumption that *before* and *after* compete with *while* (cf. Krifka 2010). Under this view, *Emma was irritable before Hector built the tent* asserts “Emma was irritable sometime before Hector finished building the tent” (see the LF in (8b)) and implicates “it is not the case that Emma was irritable while Hector built the tent.” In other words, the literal meaning of telic *before*-clauses is before-finish, on this view, but these clauses receive a stronger, before-start interpretation whenever a “not *while*” implicature is computed.

Although this implicature approach seems promising, it actually does not predict the observed asymmetry between, e.g., no-coercion and completive trials in Experiment 2. To see this, consider Contexts A and B from Table 2 again, recalling that Hector built the tent 4-4:30pm in both contexts and that the contexts differed as to when Emma was irritable (Context A, 2pm-4:30pm; Context B, 4:15-4:30pm). As discussed above, Rett (2020) predicts that *Emma was irritable before Hector built the tent* should be harder to process in Context B than in Context A, since the only way to make the sentence true in Context B (according to her theory) is to coerce. The implicature approach, by contrast, does not draw a distinction between the two contexts. In both contexts, the literal meaning of the sentence is verified; furthermore, since both contexts involve temporal overlap, they both

falsify the putative implicature $\neg$ *Emma was irritable while Hector built the tent*. Hence, the implicature approach does not straightforwardly predict an asymmetry between completive and no-coercion trials.

8.2 A non-coercive theory of atelic *within*-modifier sentences

As discussed above, the results of Experiments 1a and 3 chafe against INCHOAT-based theories of sentences like (32a) and thus raise the question of what a non-coercive theory of (32a) would look like. In this subsection, we briefly sketch a non-coercive theory of this sort, building on intuitions from Krifka (2010).

- (32) Context: Ben was exposed to a virus at 6pm.
 a. Within an hour, Ben was sick.

On the theory in question, (32a) receives the LF in (33a), ignoring tense for simplicity. In this LF, *within an hour* is the sister of a free variable t , which denotes the time-point that *within an hour* implicitly makes reference to. In context (32), for example, t is 6pm (the time of Ben's exposure). Armed with the LF in (33a), we can assume that *within an hour* has the simple entry in (33b); note that $[t, t+1\text{hour}]$ denotes the interval that starts at t and ends 1 hour after t .

- (33) a. $[[[\text{within an hour}] t][\text{Ben be sick}]]$
 b. $[[\text{within an hour}]] = \lambda t. \lambda p. \exists t' \sqsubseteq [t, t+1\text{hour}]: p(t') = 1$

In LF (33a), *within an hour* takes t , takes *Ben be sick*, and returns true iff the interval $[t, t+1\text{hour}]$ contains a time at which Ben was sick. For example, if t is 6pm, (33a) comes out true iff Ben was sick at some point in the interval $[6\text{pm}, 7\text{pm}]$.

- (34) $[[\text{Within an hour, Ben was sick}]] = 1$ iff $\exists t' \sqsubseteq [6\text{pm}, 7\text{pm}]: \text{Ben is sick at } t'$

As it stands, (34) does not quite capture the meaning of (32a). For (32a) to be true, it is not enough for the conditions in (34) to be met; additionally, Ben must have started feeling sick after 6pm. To capture this additional meaning component, we assume (following Krifka 2010) that (32a) competes with relevant alternatives where *an hour* is replaced by another time-span, e.g. (35a-b). The exact number of relevant alternatives to *within an hour* presumably depends on the context.

- (35) a. Within forty-five minutes, Ben was sick.
 b. Within half an hour, Ben was sick.

Under this theory, *within t', Ben was sick* is stronger than *within t'', Ben was sick* whenever t' is a shorter time-span than t'' . For example, in context (32), (35a) says that Ben was sick sometime during [6pm,6:45pm], whereas (32a), as we have seen, says that Ben was sick sometime during [6pm,7pm]. As a result of this strength relationship, asserting (32a) implicates the negation of (35a) in a context where (35a) is relevant. Together, (32a) and the negation of (35a) entail that there was an event of Ben becoming sick during [6pm,7pm], mirroring intuitions.

Note that this theory derives the correct meaning for telic *within*-modifier sentences like (36a) as well. On our theory, (36a) asserts that there is a complete Hector-eating-the-burrito event in the interval [6pm,6:30pm] and implicates that there is not a complete Hector-eating-the-burrito event in a shorter timespan.

- (36) Context: Hector bought a burrito at 6pm.
 a. Within half an hour, Hector ate the burrito.

Although this analysis is only a sketch, it shows that one can account for *within*-modifiers without predicting that atelic *within*-modifiers will be uniquely hard to process. In particular, the above theory does not predict that atelic *within*-modifier sentences will be more costly than analogous *for*-modifier sentences, as processing both kinds of sentences involves computing similar implicatures (*within an hour* implicates, e.g., *not within thirty minutes*; *for two hours* implicates, e.g., *not for three hours*). Hence, the theory sketched in this subsection is more compatible with the results of Exps. 1a and 3 than an INCHOAT-based theory.

8.3 Condoravdi (2010)

As discussed in section 8.1, our results are more compatible with Rett's (2020) ambiguity theory than with under-specification theories. However, there is an ambiguity theory of *after* other than Rett 2020 that we have not yet discussed: namely, that of Condoravdi (2010). In this subsection, we briefly summarize Condoravdi's (2010) theory of *after* and then assess whether her theory can account for our results.

There are two main differences between Condoravdi's (2010) theory of *after* and Rett's. First, for Condoravdi (2010), both readings of *after* involve coercion, not just the after-start reading. Additionally, Condoravdi (2010) posits that coercion in *after*-clauses is performed to resolve a type mismatch, whereas coercion for Rett (2020) must be thought of as having a pragmatic source (see section 3). To see this, consider our LF for (1a), repeated below, in light of the (slightly simplified) Condoravdi (2010)-style entry for *after* in (37b). We continue to assume that propositions are of type $\langle i, t \rangle$ and ignore tense for simplicity.

- (37) a. [Emma be happy] [after [Hector be happy]]
 b. $\llbracket \text{after} \rrbracket = \lambda t. \lambda p. \exists t' [p(t') = 1 \text{ and } t' > t]$

While Rett’s (2020) *after* requires a proposition as its first argument and can thus combine smoothly with its sister, Condoravdi’s (2010) *after* requires a time-interval (type *i*) as its first argument and thus clashes with its sister.

Condoravdi (2010) proposes that the type mismatch between (37b) and its sister gets resolved via the insertion of one of two coercion operators. The first is an operator she calls EARLIEST, which is like Rett’s INCHOAT except for the fact that it outputs a time (type *i*) rather than a singleton set of times (type $\langle i, t \rangle$). For example, if Hector was happy from 6pm to 8pm, then $\text{INCHOAT}(\llbracket \text{Hector be happy} \rrbracket) = \{6\text{pm}\}$, while $\text{EARLIEST}(\llbracket \text{Hector be happy} \rrbracket) = 6\text{pm}$. Since the latter is a time-interval, it can serve as (37b)’s first argument.

In addition to EARLIEST, Condoravdi (2010) claims that an operator MAX can also coerce $\llbracket \text{Hector be happy} \rrbracket$ into a an interval, namely the maximal interval throughout which Hector was happy. For example, if Hector was happy from 6pm-8pm, then $\text{MAX}(\llbracket \text{Hector be happy} \rrbracket) = [6\text{pm}, 8\text{pm}]$.

To summarize, in our example where Hector was happy from 6pm-8pm, *Emma was happy after Hector was happy* is predicted to have two readings, one in which EARLIEST coerces the *after*-clause and one in which MAX does. As shown in (38), the LF with EARLIEST expresses an after-start reading, whereas the LF with MAX expresses an after-finish reading.

- (38) a. $\llbracket [\text{Emma be happy}] [\text{after} [\text{EARLIEST} [\text{Hector be happy}]]] \rrbracket = 1$ iff
 $\exists t' [\text{Emma is happy at } t' \text{ and } t' > 6\text{pm}]$
 b. $\llbracket [\text{Emma be happy}] [\text{after} [\text{MAX} [\text{Hector be happy}]]] \rrbracket = 1$ iff
 $\exists t' [\text{Emma is happy at } t' \text{ and } t' > [6\text{pm}, 8\text{pm}]]$

Since both readings of atelic *after*-clauses involve coercion for Condoravdi (2010), she does not straightforwardly predict any online or offline asymmetry between the two readings, contrary to the results of Experiment 4. To account for the results of Experiment 4, Condoravdi (2010) could try to say that coercion with EARLIEST is costlier than coercion with MAX, but this claim is not obvious and would require independent motivation (which does not exist, as far as we know). Note that even if coercion with EARLIEST can be independently shown to be costlier than coercion with MAX in some way, Condoravdi’s timing predictions would not be different from Rett’s: parsers should know that the costlier coercion operator, EARLIEST, is required at the main verb in the *after*-clause, so that is where we should see a slowdown. Thus, Condoravdi’s (2010) theory is at best on the same level as Rett’s theory when it comes to explaining the results of Experiment 4.²²

²²Regarding telic *before*-clauses, Condoravdi (2010) claims that they have just one LF, the one with EARLIEST. LFs with MAX are, she claims, ill-formed when the temporal clause is telic (or non-cumulative more broadly, see p. 902). To capture the full range of readings for telic *before*-clauses, she claims that $\text{EARLIEST}(p)$ picks out the telos when p is telic, rather than the

8.4 Directions for future research

The most important direction for future research is to ascertain why the slowdowns incurred by after-start readings of *after* and before-finish readings of *before* were delayed relative to other coercions. To this end, it would be useful to run further experiments on *before-/after*-clauses that measure processing times using a paradigm that is less prone to spillover effects than self-paced reading, such as eye-tracking or the Maze task (Witzel et al. 2012). If the effects that were delayed in Experiments 2 and 4 show up in the predicted region once we switch from self-paced reading to, e.g., eye-tracking, then we could conclude that the delay in Experiments 2 and 4 was an experimental artifact. If the effects remain delayed, then the delay was probably not an experimental artifact. In addition to this direction for future experimental research, future theoretical research should continue to probe the extent to which theories other than Rett (2020) can account for our results in a non-stipulative way.

Although further research is needed, the experiments reported here suggest that there are online and offline asymmetries between the two readings of atelic *after*-clauses and between the two readings of telic *before*-clauses, constraining the range of theories of these temporal connectives.

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start-point. Hence, her theory of *before* is essentially an under-specification theory (in the sense that telic *before*-clauses have just one LF, an LF with a before-finish meaning).

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